

FLEXIBLE ALGEBRAIC CURVES AND FAKE VANISHING CYCLES

JOSHUA LEHMAN AND TUDUR LEWIS

ABSTRACT. We prove that complex curves in sufficiently ample linear systems on simply connected projective surfaces are topologically (spin) flexible: any (spin) mapping class of the underlying real surface extends to a topological mapping class of the ambient 4-manifold. As a corollary, we construct fake vanishing cycles, that is, simple closed curves on the underlying real surface that are not vanishing cycles, yet still bound a topologically embedded (± 1) -framed disk in the surface exterior. We also give examples of pushed in Seifert surfaces in D^4 that are not topologically spin-flexible, but become spin-flexible after a single internal stabilization. Our methods utilize the equivariant intersection form on the surface exterior.

1. INTRODUCTION

Fix a properly embedded oriented surface Σ in a compact oriented 4-manifold M . Consider the group $\pi_0(\text{Homeo}_\partial(M, \Sigma))$ of isotopy classes of orientation preserving homeomorphisms of M which fix Σ setwise and ∂M pointwise. Restriction determines a homomorphism

$$\mathcal{R}: \pi_0(\text{Homeo}_\partial(M, \Sigma)) \rightarrow \pi_0(\text{Homeo}_\partial(\Sigma)).$$

The index-2 subgroup $\text{Mod}(\Sigma) \leq \pi_0(\text{Homeo}_\partial(\Sigma))$ preserving the orientation of Σ is the *mapping class group* of Σ . The *topological extendable subgroup* $\mathcal{E}^{\text{TOP}}(M, \Sigma)$ of Σ in M is the image of $\mathcal{R}$ in $\text{Mod}(\Sigma)$. The surface Σ is *topologically flexible* in M if $\mathcal{E}^{\text{TOP}}(M, \Sigma) = \text{Mod}(\Sigma)$. If $\Sigma \subset M$ is characteristic and $H_1(M; \mathbb{Z}) = 0$, Rochlin defined a quadratic form $q_\Sigma: H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ which obstructs flexibility:

$$\mathcal{E}^{\text{TOP}}(M, \Sigma) \leq \text{Mod}(\Sigma)[q_\Sigma]$$

where $\text{Mod}(\Sigma)[q_\Sigma]$ denotes the stabilizer of q_Σ in $\text{Mod}(\Sigma)$. The surface Σ is *topologically spin-flexible* in M if $\mathcal{E}^{\text{TOP}}(M, \Sigma) = \text{Mod}(\Sigma)[q_\Sigma]$.

We find sufficient conditions under which complex curves in simply connected projective surfaces are topologically (spin) flexible.

Let X be a projective surface and $L \in \text{Pic}(X)$ a line bundle¹. Recall that holomorphic sections of L , elements of $H^0(X; L)$, have the same zero locus in X if and only if they are scalar multiples of one another [GH94, 1.4]. Therefore elements of $|L| := \mathbb{P}H^0(X; L)$ are identified with complex curves in X . A source of extendable mapping classes for a smooth curve $C \in |L| := \mathbb{P}H^0(X; L)$ is the *monodromy* $\Gamma_L \leq \text{Mod}(C)$ of the linear system $|L|$: a homotopy class of a loop in the smooth locus $U_L \subset |L|$ based at C produces a loop of smooth curves in X , and in particular, a mapping class of C . By isotopy extension,

$$\Gamma_L \leq \mathcal{E}^{\text{TOP}}(X, C).$$

For sufficiently ample line bundles, our work establishes a difference between these two groups. Recall that L is *very ample* (or *1-jet ample*) if there is a holomorphic embedding $\varphi: X \rightarrow \mathbb{C}P^N$ such that $L = \varphi^*H$ where H is the dual of the tautological line bundle on $\mathbb{C}P^N$ [GH94, 1.4]. [BS93] introduced a quantitative refinement, known as *k-jet ampleness*, with the key property that if L_1 is k_1 -jet ample and L_2 is k_2 -jet ample, $L_1 \otimes L_2$ is $(k_1 + k_2)$ -jet ample (Proposition 6.5)

Theorem A. *Let X be a projective surface with $\pi_1(X) = 1$. Let $L \in \text{Pic}(X)$. Suppose that L admits a decomposition $L = L_1 \otimes L_2$ for L_1 very ample and L_2 6-jet ample. Then smooth curves in the linear system $\mathbb{P}H^0(X; L)$ are topologically (spin) flexible.*

Corollary B. *Smooth plane curves in $\mathbb{C}P^2$ are topologically (spin) flexible. Smooth curves in $\mathbb{C}P^1 \times \mathbb{C}P^1$ of bidegree (d_1, d_2) with $d_1, d_2 \geq 6$ are topologically (spin) flexible.*

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¹Recall that $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ is the group of isomorphism classes of holomorphic line bundles on X , and, as $\pi_1(X) = 1$, the first Chern class gives an injective homomorphism $c_1: \text{Pic}(X) \rightarrow H^2(X; \mathbb{Z})$ [GH94, 1.1]

Hirose proved that cubics and quartics in $\mathbb{CP}^2$ are (smoothly) flexible [Hir05]. Work of Salter implies (smooth) spin-flexibility of quintics [Sal25]. Whilst Corollary B establishes topological (spin) flexibility of plane curves in all remaining degrees, their *smooth flexibility* (Definition 2.1) remains open. We note that after a single *internal stabilization*, plane curves become smoothly (spin) flexible [Leh26].

Theorem A follows from a more technical version (Theorem 6.8) and work of Banerjee-Salter [BS25]. Precisely, Theorem 6.8 requires that $L \in \text{Pic}(X)$ is *6-jet ample* and the linear system $|L|$ has *r-spin monodromy* (Definition 6.4). Banerjee-Salter prove that the line bundles considered in Theorem A satisfy the *r-spin monodromy condition* [BS25, Theorem A].

Recent work of Baykur and Sunukjian [BS26], in conjunction with Corollary B, establishes the existence of surfaces in $\mathbb{CP}^2$ that are topologically isotopic to plane curves, so are topologically flexible, but are *projectively rigid* in the smooth category: the image of their smooth extendable subgroup under the *projective symplectic representation* is trivial [BS26, Remark 19].

Fake vanishing cycles. A *vanishing cycle* on a smooth complex curve $C \in |L|$ is a simple closed curve on C which contracts to a single point under a *nodal degeneration* of C in X . Under this degeneration, the curve traces out a (-1) -framed disk in X with interior disjoint from C . Moreover, the monodromy of this degeneration can be used to exhibit the *Dehn twist* along the vanishing cycle as an element in Γ_L , and in particular, as a (smoothly) extendable mapping class. For details, we refer the reader to [AGZV12]. We call a simple closed curve on C a *fake vanishing cycle* if it is not a vanishing cycle on C , but still bounds a topologically embedded (± 1) -framed disk in X with interior disjoint from C . By Proposition 2.7, the Dehn twist on any fake vanishing cycle is a topologically extendable mapping class. Let $K_X \in \text{Pic}(X)$ be the *canonical bundle* of X (the top exterior power of the holomorphic cotangent bundle of X).

Corollary C. *Let X be a projective surface with $\pi_1(X) = 1$. Let $L \in \text{Pic}(X)$ satisfy the hypothesis of Theorem A. Fix a smooth curve $C \in |L|$. If the maximal root of the adjoint line bundle $K_X \otimes L$ in $\text{Pic}(X)$ has order at least three, then there exists a fake vanishing cycle $a \subset C$.*

Remark 1.1 (Fake vanishing cycles don't always exist). Recall that the *hyperplane bundle* $\mathcal{O}(1) \rightarrow \mathbb{CP}^2$ is the dual of the tautological line bundle on $\mathbb{CP}^2$. Let $C_d \subset \mathbb{CP}^2$ be a smooth curve of degree d . Then C_d may be identified as the zero locus of a section of $\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$ [GH94, Chapter 1, Section 3]. The adjoint line bundle of $\mathcal{O}(d)$ is $K_{\mathbb{CP}^2} \otimes \mathcal{O}(d) \cong \mathcal{O}(d-3) \cong \mathcal{O}(1)^{\otimes(d-3)}$. It follows from Corollary C that complex curves of degree $d \geq 6$ have fake vanishing cycles. On the other hand, complex curves of degree $d \leq 5$ do not. This is due to the fact that the monodromy $\Gamma_{\mathcal{O}(d)}$ (Definition 6.1) of the associated linear system is either the entire mapping class group ($d = 3, 4$), or the stabilizer of Rochlin's form ($d = 5$). See Remark 6.13 for more details.

For a smooth curve $C \in |L|$, our approach to Theorem A uses work of Conway-Powell [CP23] to produce homeomorphisms of the pair (X, C) . For surfaces with knot group $\mathbb{Z}$, [CP23] implicitly gives a complete algebraic criterion for extending mapping classes topologically. This is an application of [CP23, Theorem 4.1] to the surface exterior (recorded here as Theorem 4.6). We implement this by finding a subsurface $F \subset C$ contained in a 4-ball $D^4 \subset X$ with local knot group $\pi_1(D^4 - F) \cong \mathbb{Z}$. The surface F is a *pushed in Seifert surface*.

Remark 1.2. Recent work of Conway and Kasprowski [CK26, Theorem 1.12] gives a complete algebraic characterization of the topological extendable subgroup whenever the knot group of the surface is finite cyclic. For a smooth degree d plane curve $C_d \subset \mathbb{CP}^2$, we have $\pi_1(\mathbb{CP}^2 - C_d) \cong \mathbb{Z}_d$. Whilst it is unclear to the authors if the computation [CK26, Theorem 1.12] immediately implies the topological (spin) flexibility of C_d , we note that it gives a direct approach to constructing homeomorphisms of the pair $(\mathbb{CP}^2, C_d)$. In this paper we use [CP23, Theorem 4.1] to extend homeomorphisms, and the analogous criterion for finite cyclic knot groups is [CK26, Theorem 1.6].

Obstructing extendable mapping classes. If $F \subset S^3 = \partial D^4$ is a Seifert surface, consider the properly embedded surface (again denoted) $F \subset D^4$ obtained by pushing the interior of F into the interior of D^4 . We refer to such a surface as a *pushed in Seifert surface*. Let $T \subset S^4$ denote an unknotted torus. A *trivial internal stabilization* of $F \subset D^4$ is the relative connected sum $(D^4, F) \# (S^4, T)$.

Theorem D. *There exist infinitely many non-isotopic pushed in Seifert surfaces in D^4 that are not topologically spin-flexible rel boundary, but become topologically spin-flexible after a trivial internal stabilization.*

The obstruction in Theorem D arises from (a specialization of the) *equivariant intersection form* on the exterior of F in D^4 .

Remark 1.3. Several approaches to obstructing extendable mapping classes in the literature have exploited the *peripheral structure*² associated to an embedding $\Sigma \subset M$. For example, work of Hirose [Hir93] used the peripheral structure to compute the extendable subgroup of (turned) spun tori in S^4 . Recently, Niu [Niu26] used the peripheral subgroup and iterated rim surgery to produce closed surfaces in any 4-manifold with *trivial* topological extendable subgroup (a topologically *rigid* surface). Note that [BS26] had previously constructed a topologically rigid surface using hyperbolic geometry. The peripheral subgroup of pushed in Seifert surfaces is too simple to obstruct extendable mapping classes, indeed it is $\mathbb{Z}$, generated by a meridian of the surface.

Problem 4.39 on the K3 list [BKR26] asks which subgroups of $\text{Mod}(\Sigma)$ arise as extendable subgroups. A linearization of this problem asks which subgroups of $\text{Sp}_{2g}(\mathbb{Z}/2\mathbb{Z})$ arise as the image of extendable subgroups under the mod 2 symplectic representation (see [Kaw07]). We show that *mod 2 Weyl groups* of type A and E may be realized.

The *Milnor fiber* $F_\Gamma \subset D^4$ for the link L of a singularity of type ADE with Dynkin diagram Γ is, up to isotopy, a pushed in Seifert surface. We equip the surface F_Γ with an ordered configuration $\mathcal{C}_\Gamma$ of curves satisfying the following properties: the homology classes of curves in $\mathcal{C}_\Gamma$ generate $H_1(F_\Gamma; \mathbb{Z}/2\mathbb{Z})$, the *intersection graph* associated to $\mathcal{C}_\Gamma$ is Γ , and if S is the Seifert form of F_Γ relative to $\mathcal{C}_\Gamma$, then $S + S^T$ is the *Cartan matrix* of Γ .

Theorem E. *Let Γ be a Coxeter–Dynkin diagram of type A_n with n even, E_6 or E_8 . Let $W_2(\Gamma)$ be the subgroup of $\text{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ generated by the transvections in the homology classes of the curves in $\mathcal{C}_\Gamma$. This is the mod 2 reduction of the Weyl group of type Γ . Then*

$$\Psi_2(\mathcal{E}^{\text{TOP}}(D^4, F_\Gamma)) = W_2(\Gamma)$$

where $\Psi_2 : \text{Mod}(F_\Gamma) \rightarrow \text{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ is the mod 2 symplectic representation.

Remark 1.4. The singularities D_n , A_n for n odd and E_7 are excluded from Theorem E because our methods rely on the associated Seifert surface having boundary a *knot* in S^3 , rather than a link of more than one component.

1.1. General overview.

1.1.1. *Theorem A/Theorem 6.8.* We begin by finding a 4-ball $D^4 \subset X$ and a smooth curve $C \in |L|$ such that $F = C \cap D^4$ has knot group $\pi_1(D^4 - F) \cong \mathbb{Z}$ and sufficiently large genus (the surface F is a pushed in Seifert surface of genus three). This is where the 6-jet ampleness of L is used (Lemma 6.6). We then use results of Conway–Powell [CP23] to upgrade isometries of the equivariant intersection form on the exterior of F in D^4 to homeomorphisms of the pair (D^4, F) rel boundary. In particular, we obtain that the *Torelli subgroup* of $\text{Mod}(F)$ is topologically extendable:

$$\mathcal{I}(F) \leq \mathcal{E}^{\text{TOP}}(D^4, F).$$

These homeomorphisms may be extended by the identity to the rest of X , yielding extendable mapping classes for the pair (X, C) . By assumption (Theorem 6.8), the linear system $|L|$ has r -spin monodromy (Definition 6.4). This implies that $\mathcal{E}^{\text{TOP}}(X, C)$ contains an r -spin mapping class group (Corollary 6.2). We conclude by exploiting how r -spin mapping class groups intersect this Torelli subgroup.

²If $\nu\Sigma$ is a closed tubular neighbourhood of Σ in M , the *peripheral subgroup* of Σ in M is the image of the map $\iota_* : \pi_1(\partial\nu\Sigma) \rightarrow \pi_1(M - \Sigma)$ induced by inclusion.

1.1.2. *Theorem D and E.* The obstruction in Theorem E is a consequence of the fact that homeomorphisms of the pair (D^4, F) can be promoted to isometries of the intersection form of the 2-fold branched cover $\Sigma_2(D^4, F)$ of D^4 in F . We realize this as a specialization of the equivariant intersection form on the exterior of F in D^4 (Theorem 5.1). The Seifert surfaces associated to the A_n -family for n even are the family used in Theorem D. Their flexibility after a single stabilization is a computation using basic properties of r -spin mapping class groups.

1.2. **Outline of paper.** Section 2 defines the extendable subgroup. Section 3 recalls the necessary background on r -spin mapping class groups and discusses how r -spin mapping class groups intersect the Torelli subgroup (Lemma 3.11). Section 4 begins by recalling techniques for extending homeomorphisms of the boundary of a 4-manifold with fundamental group $\mathbb{Z}$ due to Conway-Powell [CP23] (Theorem 4.6). From this we extract methods for extending mapping classes (Corollary 4.9). We give an explicit treatment of this procedure for pushed in Seifert surfaces in D^4 , illustrated by generalizing classical methods (Proposition 2.7) for extending mapping classes in the smooth category (Proposition 4.20). Section 5 proves Theorems D and E. Section 6 proves Theorem A.

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2. EXTENDABLE MAPPING CLASSES AND ROCHLIN'S FORM

Let M be a compact, oriented, smooth 4-manifold. Let $\Sigma \subset M$ be a smooth, compact, properly embedded, oriented surface in M .

2.1. **Extendable subgroups.** The *mapping class group* of Σ is the group $\text{Mod}(\Sigma) = \pi_0(\text{Homeo}_\partial^+(\Sigma))$ of isotopy classes of orientation preserving homeomorphisms of Σ that fix $\partial\Sigma$ pointwise. Let $\pi_0(\text{Homeo}_\partial(M, \Sigma))$ (resp. $\pi_0(\text{Diff}_\partial(M, \Sigma))$) denote the group of isotopy classes of orientation preserving homeomorphisms (resp. diffeomorphisms) of M which fix Σ setwise and ∂M pointwise. Restriction determines a homomorphism $\mathcal{R}^{\text{TOP}}: \pi_0(\text{Homeo}_\partial(M, \Sigma)) \rightarrow \pi_0(\text{Homeo}_\partial(\Sigma))$ (resp. $\mathcal{R}^{\text{DIFF}}: \pi_0(\text{Diff}_\partial(M, \Sigma)) \rightarrow \pi_0(\text{Homeo}_\partial(\Sigma))$).

Definition 2.1. The *topological* (resp. *smooth*) *extendable subgroup* $\mathcal{E}^{\text{TOP}}(M, \Sigma)$ (resp. $\mathcal{E}^{\text{DIFF}}(M, \Sigma)$) of Σ in M is the image of $\mathcal{R}^{\text{TOP}}$ (resp. $\mathcal{R}^{\text{DIFF}}$) in $\text{Mod}(\Sigma)$. We say that Σ is *topologically* (resp. *smoothly*) *flexible* if $\mathcal{E}^{\text{TOP}}(M, \Sigma) = \text{Mod}(\Sigma)$ (resp. $\mathcal{E}^{\text{DIFF}}(M, \Sigma) = \text{Mod}(\Sigma)$).

If the surface $\Sigma \subset M$ is characteristic, that is, the second Stiefel-Whitney class $w_2(M)$ is the mod 2 reduction of $\text{PD}(\Sigma)$, there is an obstruction to topological flexibility.

2.2. **Rochlin's form and spin-flexibility.** If the surface $\Sigma \subset M$ is characteristic, that is, the second Stiefel-Whitney class $w_2(M)$ is the mod 2 reduction of $\text{PD}(\Sigma)$, there is an obstruction to topological flexibility. For a properly embedded, characteristic surface $\Sigma \subset M$ in a compact 4-manifold M with $H_1(M; \mathbb{Z}) = 0$, *Rochlin's quadratic form* $q_\Sigma: H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ is defined. If $\gamma \subset \Sigma$ is a simple closed curve, choose a (locally flat) surface $S \subset M$ with $\partial S = \gamma$, and interior *topologically transverse* to Σ . Then

$$q_\Sigma([\gamma]) = |\Sigma \cap \text{int}(S)| + e_\Sigma(S) \pmod{2}$$

where $e_\Sigma(S)$ is the *relative Euler number* of S (the integer valued obstruction to extending the surface framing of ∂S induced by Σ to all of S). We refer the reader to [Klu21, Mat86] for a precise definitions and proof.

Remark 2.2. The proof in [Mat86, Section 5] is carried out in the smooth category. However, the techniques go through in the topological category, due to the existence of topological normal bundles and topological transversality.

The mapping class group $\text{Mod}(\Sigma)$ acts on the set of quadratic forms on $H_1(\Sigma; \mathbb{Z}/2\mathbb{Z})$ by pullback,

$$(f \cdot q)(x) = q(f_*^{-1}(x))$$

for $f \in \text{Mod}(\Sigma)$ and $q: H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$.

Definition 2.3. For a quadratic form $q : H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$, let

$$\text{Mod}(\Sigma)[q] = \text{Stab}_{\text{Mod}(\Sigma)}(q).$$

Given the definition of q_Σ , we obtain the following result (which requires only that M be a topological manifold).

Proposition 2.4. *Fix a properly embedded characteristic surface Σ in a 4-manifold M . Assume $H_1(M; \mathbb{Z}) = 0$. Then*

$$\mathcal{E}^{\text{TOP}}(M, \Sigma) \leq \text{Mod}(\Sigma)[q_\Sigma]$$

where $q_\Sigma : H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ denotes Rochlin's quadratic form.

Definition 2.5. A properly embedded characteristic surface $\Sigma \subset M$ is *topologically* (resp. *smoothly*) *spin-flexible* in M if $\mathcal{E}^{\text{TOP}}(M, \Sigma) = \text{Mod}(\Sigma)[q_\Sigma]$ (resp. $\mathcal{E}^{\text{DIFF}}(M, \Sigma) = \text{Mod}(\Sigma)[q_\Sigma]$).

2.3. Some extendable mapping classes. In the sequel, we use the following methods for extending mapping classes. Montesinos proved that an unknotted torus $\Sigma_1 \subset S^4$ is smoothly spin-flexible [Mon83]. Hirose proved that unknots $\Sigma_g \subset S^4$ of all genera are smoothly spin-flexible [Hir02]. These statements have rel boundary versions. Let $\Sigma_g^1 \subset D^4$ be the result of removing a 4-ball from the pair (S^4, Σ_g) meeting Σ_g in a closed 2-disk.

Proposition 2.6. *For $g \geq 1$, $\mathcal{E}^{\text{DIFF}}(D^4, \Sigma_g^1) = \text{Mod}(\Sigma_g^1)[q_{\Sigma_g^1}]$.*

Proof. This follows from the statements in the closed case. For details we refer the reader to [Leh26, Proposition 2.15]. $\square$

For a simple closed curve $\gamma \subset \Sigma$, let T_γ be the (left-handed) Dehn twist about γ .

Proposition 2.7 (Hopf band trick). *Let $\gamma \subset \Sigma \subset M$ be a simple closed curve. Assume that γ bounds a topologically (resp. smoothly) embedded disk $D \subset \text{int}(M)$ with interior disjoint from Σ and $e_\Sigma(D) = \pm 1$. Then $T_\gamma \in \mathcal{E}^{\text{TOP}}(M, \Sigma)$ (resp. $T_\gamma \in \mathcal{E}^{\text{DIFF}}(M, \Sigma)$).*

Proof. See the proof of [LNSW13, Lemma 6.2]. $\square$

Remark 2.8. In the setting of Proposition 2.7, let $\nu_{\gamma \subset \Sigma} \cong S^1 \times [0, 1]$ be a closed tubular neighbourhood of γ in Σ . Let $\nu_D \cong D \times D^2$ be a closed tubular neighbourhood of D in M . Up to isotopy, $\nu_{\gamma \subset \Sigma}$ is a *Hopf band* in $S^3 = \partial(D \times D^2)$, that is, an annulus with ± 1 full twist (computed as the linking number of the boundary components in S^3) and unknotted core.

3. BACKGROUND ON r -SPIN MAPPING CLASS GROUPS

We recall fundamental results about stabilizers of r -spin structures on closed surfaces (r -spin mapping class groups). We conclude this section by exploiting how these stabilizers intersect the Torelli subgroup if $r > 2$. This section follows [Sal19] and [CS21] closely.

3.1. Higher spin structures and winding number functions. Let Σ be a closed, oriented surface. Let $UT\Sigma \rightarrow \Sigma$ denote the unit tangent bundle of Σ and $\langle -, - \rangle$ the oriented intersection pairing on $H_1(\Sigma; \mathbb{Z})$.

Definition 3.1. An r -spin structure on Σ is a cohomology class $\xi \in H^1(UT\Sigma; \mathbb{Z}/r\mathbb{Z})$ where $\xi(\mathbf{z}) = 1$ for $\mathbf{z}$ a positively oriented fiber of $UT\Sigma \rightarrow \Sigma$.

Let $\mathcal{S}(\Sigma)$ denote the set of oriented isotopy classes of simple closed curves on Σ .

Definition 3.2. An $\mathbb{Z}/r\mathbb{Z}$ -winding number function on Σ is a function $\psi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$ satisfying

(a) (*Twist-linearity*) For any $a, b \in \mathcal{S}(\Sigma)$,

$$\psi(T_a(b)) = \psi(b) + \langle b, a \rangle \psi(a) \pmod{r}.$$

(b) (*Homological coherence*) Let $S \subset \Sigma$ be a subsurface with boundary components $c_1, \dots, c_k$ oriented with S to their left. Then

$$\sum_{i=1}^k \psi(c_i) = \chi(S) \pmod{r}$$

where $\chi(S)$ denotes the Euler characteristic of S .

Remark 3.3. By homological coherence, $r \mid \chi(\Sigma)$. If $\bar{a}$ denotes $a \in \mathcal{S}(\Sigma)$ with the opposite orientation, then $\psi(\bar{a}) = -\psi(a) \pmod{r}$.

Example 3.4 (Constructing winding number functions). Fix an r -spin structure $\xi \in H^1(UT\Sigma; \mathbb{Z}/r\mathbb{Z})$. Given $a \in \mathcal{S}(\Sigma)$, let $\vec{a} \subset UT\Sigma$ denote the unit forward pointing velocity vector field of a . The assignment $a \rightarrow \vec{a}$ is referred to as the *Johnson lift* [Joh80]. Define a function $\phi_\xi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$ by $\phi_\xi(a) = \xi(\vec{a})$.

Theorem 3.5. *The assignment*

$$\xi \in H^1(UT\Sigma; \mathbb{Z}/r\mathbb{Z}) \rightarrow \phi_\xi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$$

(*Example 3.4*) defines a bijective correspondence between r -spin structures on Σ and mod r winding number functions on Σ . In particular, there are r^{2g} winding number functions on Σ valued in $\mathbb{Z}/r\mathbb{Z}$.

Proof. See [HJ89] and e.g. [Sal25, Sections 2, 3 and Theorem 3.2]. $\square$

3.2. r -spin mapping class groups. The mapping class group $\text{Mod}(\Sigma)$ acts on the set of $\mathbb{Z}/r\mathbb{Z}$ valued winding number functions by pullback. Precisely

$$(f \cdot \psi)(a) = \psi(f^{-1}(a))$$

for $f \in \text{Mod}(\Sigma)$ and $a \in \mathcal{S}(\Sigma)$.

Definition 3.6. Let $\psi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$ be a winding number function on Σ . The stabilizer of ψ in $\text{Mod}(\Sigma)$ is denoted

$$\text{Mod}(\Sigma)[\psi] = \{f \in \text{Mod}(\Sigma) \mid f \cdot \psi = \psi\}.$$

We call $\text{Mod}(\Sigma)[\psi]$ an *r -spin mapping class group*.

Definition 3.7. A simple closed curve $a \subset \Sigma$ is *admissible* under a winding number function $\psi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$ if it is nonseparating and $\psi(a) = 0$.

By twist-linearity, if $a \subset \Sigma$ is ψ -admissible, $T_a \in \text{Mod}(\Sigma)[\psi]$. Calderon-Salter [CS21] established that ψ -admissible twists are enough to generate the r -spin mapping class group $\text{Mod}(\Sigma)[\psi]$.

Theorem 3.8 ([CS21, Proposition 5.1]). *Let $\psi : \mathcal{S}(\Sigma_g) \rightarrow \mathbb{Z}/r\mathbb{Z}$ be a winding number function on a closed surface Σ_g of genus $g \geq 3$. Then*

$$\text{Mod}(\Sigma_g)[\psi] = \langle T_a \mid a \subset \Sigma_g, a \text{ admissible} \rangle.$$

We require two instances of the *r -spin change of coordinates* principle. We refer the reader to [CS21, Section 5.2] for more details. Let Σ_g be a closed oriented surface of genus g and fix a winding number function $\psi : \mathcal{S}(\Sigma_g) \rightarrow \mathbb{Z}/r\mathbb{Z}$.

Proposition 3.9. *Let $a, b \subset \Sigma$ be nonseparating simple closed curves on Σ_g , $g \geq 2$. Assume that $\psi(a) = \psi(b)$. Then there exists $f \in \text{Mod}(\Sigma_g)[\psi]$ such that $f(a) = b$.*

Proof. See [CS21, Lemma 5.8]. $\square$

Proposition 3.10. *Let $a \subset \Sigma_g$ be a nonseparating simple closed curve and $g \geq 2$. Then there exists a ψ -admissible curve $b \subset \Sigma_g$ with geometric intersection number $i(a, b) = 1$.*

Proof. If r is even, this follows from [CS21, Lemma 5.7], and if r is odd, this follows from [CS21, Lemma 5.8]. $\square$

3.3. The Torelli group and r -spin mapping class groups. Let $\Psi : \text{Mod}(\Sigma) \rightarrow \text{Sp}(H_1(\Sigma; \mathbb{Z}))$ be the *symplectic representation* of $\text{Mod}(\Sigma)$. The kernel $\mathcal{I}(\Sigma)$ of Ψ is the *Torelli subgroup* of $\text{Mod}(\Sigma)$. Recall that two disjoint simple closed curves $a, b \subset \Sigma$ are a *genus h bounding pair* if they cobound some genus h subsurface on Σ . The *bounding pair map* is $T_a T_b^{-1}$ and, as a and b are homologous, $T_a T_b^{-1} \in \mathcal{I}(\Sigma)$. The following result is an essential ingredient in proving Theorem A. It exploits how r -spin mapping class groups intersect Torelli.

Lemma 3.11. *Let $\psi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/r\mathbb{Z}$ be a winding number function on Σ . Assume that Σ has genus ≥ 3 . If r is even, let $\bar{\psi}$ denote the mod 2 reduction of ψ . Let $T_a T_b^{-1} \in \text{Mod}(\Sigma)$ be a genus one bounding pair. Then*

$$\langle \text{Mod}(\Sigma)[\psi], T_a T_b^{-1} \rangle = \begin{cases} \text{Mod}(\Sigma) & \text{if } r \text{ is odd,} \\ \text{Mod}(\Sigma)[\bar{\psi}] & \text{if } r \text{ is even.} \end{cases}$$

Proof. Let $\Gamma = \langle \text{Mod}(\Sigma)[\psi], T_a T_b^{-1} \rangle$. Orient a and b so that they bound a surface of genus one to their left. Consider a simple closed curve $c \subset \Sigma$. By homological coherence, $\psi(b) + \psi(a) = -2$. Compute $\psi(T_a T_b^{-1}(c))$ using twist-linearity and homological coherence,

$$\begin{aligned} \psi(T_a T_b^{-1}(c)) &= \psi(c) - \langle c, b \rangle \psi(b) + \langle c, a \rangle \psi(a) \pmod{r} \\ &= \psi(c) + \langle c, a \rangle (\psi(b) + \psi(a)) \pmod{r} \\ &= \psi(c) - 2\langle c, a \rangle \pmod{r}. \end{aligned}$$

Fix $c \subset \Sigma$ for which $\langle c, a \rangle = 1$ and $\psi(c) = 0$ (Proposition 3.10). Then $\psi(T_a T_b^{-1}(c)) = -2$. Let $d = T_a T_b^{-1}(c)$. As $T_c \in \text{Mod}(\Sigma)[\psi]$, it follows that $T_d \in \Gamma$. Let $e \subset \Sigma$ be a simple closed curve with $\langle e, d \rangle = 1$ and $\psi(e) = 0$ (Proposition 3.10). Then

$$\psi(T_d^{-k}(e)) = 2k$$

and $T_d^{-k}(e) \in \Gamma$. Assume r is even. Using the family of curves $T_d^{-k}(e)$ and Proposition 3.9, Γ contains the Dehn twist on any nonseparating simple closed curve $\gamma \subset \Sigma$ with $\psi(\gamma) = 0 \pmod{2}$. The result now follows from Theorem 3.8.

Assume r is odd. We have shown that there exists a Dehn twist on a nonseparating curve $\gamma \subset \Sigma$ with $\psi(\gamma) = 1 \pmod{r}$ contained in Γ . Use again a geometrically dual admissible curve to show that Γ contains Dehn twists on nonseparating simple closed curves with any prescribed winding number. By Proposition 3.9, Γ contains the Dehn twist on any nonseparating simple closed curve, and hence $\Gamma = \text{Mod}(\Sigma)$. $\square$

3.4. Explicit generating sets for r -spin mapping class groups. We record a result which gives explicit finite generating sets for r -spin mapping class groups. It is used in proving Theorem D. A *network* $\mathcal{C}$ on Σ is a collection of simple closed curves on Σ such that any two curves in $\mathcal{C}$ intersect at most once, and there are no triple intersections. We say $\mathcal{C}$ is *filling* if $\Sigma \setminus \bigcup_{c \in \mathcal{C}} c$ is a collection of disks.

Definition 3.12. The *intersection graph* $\Lambda_{\mathcal{C}}$ of a network $\mathcal{C}$ on Σ is the graph with vertex set $\mathcal{C}$ and an edge between two vertices if they intersect once.

Proposition 3.13 ([CS23, Corollary 3.11]). *Let $\mathcal{C}$ be a filling network of curves on Σ_g for $g \geq 5$. Suppose that the intersection graph $\Lambda_{\mathcal{C}}$ is a tree which contains the E_6 Dynkin diagram as a subgraph and that $\mathcal{C}$ cuts the surface into n polygons with $4(k_1 + 1), \dots, 4(k_n + 1)$ sides. Set $r = \gcd(k_1, \dots, k_n)$. Then there is a unique mod r winding number function $\psi : \mathcal{S}(\Sigma_g) \rightarrow \mathbb{Z}/r\mathbb{Z}$ for which*

$$\mathcal{T}_{\mathcal{C}} = \langle T_c : c \in \mathcal{C} \rangle = \text{Mod}(\Sigma_g)[\psi].$$

3.5. Quadratic forms and mod 2 winding number functions. A result of Johnson together with Theorem 3.5 allows us to identify the stabilizer of a quadratic form $q : H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ in $\text{Mod}(\Sigma)$ with a 2-spin mapping class group. We may then apply results from Subsection 3.1.

Theorem 3.14. *A winding number function $\psi : \mathcal{S}(\Sigma) \rightarrow \mathbb{Z}/2\mathbb{Z}$ defines a quadratic form $q_{\psi} : H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ by the requirement that*

$$q_{\psi}(\gamma) = \psi(\gamma) + 1 \pmod{2}$$

for $\gamma \in \mathcal{S}(\Sigma)$. Conversely, a quadratic form on $H_1(\Sigma; \mathbb{Z}/2\mathbb{Z})$ defines a $\mathbb{Z}/2\mathbb{Z}$ winding number function by the same formula.

Proof. This follows from [Joh80, Theorem 3A] in combination with Theorem 3.5. $\square$

In particular, we have a bijective correspondence between mod 2 winding number functions on Σ and quadratic forms on $H_1(\Sigma; \mathbb{Z}/2\mathbb{Z})$. We use the following convention when dealing with admissible curves. For a form q , a curve $a \subset \Sigma$ is *admissible* if it is admissible under the corresponding 2-spin structure, in particular, if $q([a]) = 1$.

4. EXTENDING HOMEOMORPHISMS

Let $\Sigma \subset X$ be a closed surface in a closed 4-manifold X . Then Σ admits a closed tubular neighbourhood $\nu\Sigma \subset X$ [Fre82, Theorem 9.3A]. A homeomorphism of the pair $h : (X, \Sigma) \rightarrow (X, \Sigma)$ induces a homeomorphism of the exterior $X_\Sigma = X - \nu\Sigma$ (by uniqueness of tubular neighbourhoods up to isotopy [Fre82, Theorem 9.3D]). Lifting to the universal cover of X_Σ , we obtain a (semilinear) isometry of the *equivariant intersection form* on the surface exterior X_Σ . Conway-Powell [CP23] establish a converse for $\mathbb{Z}$ -surfaces. If $\pi_1 X_\Sigma \cong \mathbb{Z}$, an isometry of the equivariant intersection form on the surface exterior can be promoted to a homeomorphism of the pair (X, Σ) . We recall the relevant machinery from [CP23] (Theorem 4.6) and applications to extending mapping classes (Corollary 4.9).

4.1. Equivariant intersection form. Let M be a connected, compact, topological³ 4-manifold with fundamental group $\pi = \pi_1(M)$ and connected boundary.

Definition 4.1. The *equivariant intersection form*

$$\lambda_M : H_2(M; \mathbb{Z}[\pi]) \times H_2(M; \mathbb{Z}[\pi]) \rightarrow \mathbb{Z}[\pi]$$

has right adjoint given by

$$H_2(M; \mathbb{Z}[\pi]) \longrightarrow H_2(M, \partial M; \mathbb{Z}[\pi]) \xrightarrow{\text{PD}} H^2(M; \mathbb{Z}[\pi]) \xrightarrow{\text{ev}} \text{Hom}_{\mathbb{Z}[\pi]}(H_2(M; \mathbb{Z}[\pi]), \mathbb{Z}[\pi]),$$

the first map induced by inclusion of pairs, the second Poincaré-Lefschetz duality with twisted coefficients, and the third evaluation (see e.g. [KPT25, Section 3.1]).

In the sequel, we make use of two basic algebraic properties that λ_M satisfies. Recall that the group ring $\mathbb{Z}[\pi]$ carries a natural involution $x \mapsto \bar{x}$ determined by extending inversion on group elements linearly.

Proposition 4.2 (c.f. discussion after [KPT25, Proposition 3.12]). *The pairing λ_M is Hermitian, that is, it satisfies $\lambda_M(x, y) = \overline{\lambda_M(y, x)}$. It is also sesquilinear, that is, $\lambda_M(px, qy) = p\lambda_M(x, y)\bar{q}$ for $p, q \in \mathbb{Z}[\pi]$.*

Definition 4.3. A homomorphism $\alpha : H_2(M; \mathbb{Z}[\pi]) \rightarrow H_2(M; \mathbb{Z}[\pi])$ of $\mathbb{Z}[\pi]$ -modules is an *isometry* of λ_M if it is an isomorphism of $\mathbb{Z}[\pi]$ -modules and $\lambda_M(\alpha(x), \alpha(y)) = \lambda_M(x, y)$ for all $x, y \in H_2(M; \mathbb{Z}[\pi])$. Let $\text{Isom}(\lambda_M)$ denote the isometry group of λ_M .

Remark 4.4. For simplicity, assume that M is smooth. The pairing $\lambda_M : H_2(M; \mathbb{Z}[\pi]) \times H_2(M; \mathbb{Z}[\pi]) \rightarrow \mathbb{Z}[\pi]$ has a geometric description, justifying its title. It is given by

$$\lambda_M(x, y) = \sum_{k \in \mathbb{Z}} \langle y, t^k x \rangle t^{-k} \in \mathbb{Z}[\pi],$$

where $\langle -, - \rangle$ denotes the ordinary algebraic intersection of cycles in the universal cover of M (see e.g. [KPT25, Section 3.1]).

³The results in this section are all in the topological category. However, we note that the 4-manifolds we consider are smoothable.

4.2. 4-manifolds with $\pi_1 \cong \mathbb{Z}$. Assume that $\pi_1 M \cong \mathbb{Z}$. Let $\Lambda = \mathbb{Z}[\pi_1 M] \cong \mathbb{Z}[t, t^{-1}]$. Assume further that M has *ribbon boundary*, that is, inclusion induces a surjection $\pi_1 \partial M \rightarrow \pi_1 M$, and that $H_1(\partial M; \Lambda)$ is a torsion Λ -module. Under these hypotheses, $H_2(M; \Lambda)$ and $H_2(M, \partial M; \Lambda)$ are free Λ -modules.

An isometry of λ_M acts on $H_1(\partial M; \Lambda)$ as follows. The evaluation map $\text{ev}: H_2(M; \Lambda) \rightarrow \text{Hom}_\Lambda(H_2(M; \Lambda), \Lambda)$ is injective (by an application of the universal coefficient spectral sequence, see e.g. [KPT25, Proposition 3.3]) and $H_2(\partial M; \Lambda) \rightarrow H_2(M; \Lambda)$ is the zero map (proof of [CP23, Lemma 3.2]). Let $\overline{H_2(M; \Lambda)^*}$ denote the dual of $H_2(M; \Lambda)$ but with Λ -module structure $p \cdot x = \bar{p}x$ for $p \in \Lambda$. The right adjoint of λ_M determines an injective Λ -linear map

$$\hat{\lambda}_M : H_2(M; \Lambda) \rightarrow \overline{H_2(M; \Lambda)^*}$$

defined by $\hat{\lambda}_M(y) = \lambda_M(-, y)$. Inverting the dual of an element $\alpha \in \text{Isom}(\lambda_M)$ induces an isomorphism

$$[\alpha^{-*}] : \text{coker}(\hat{\lambda}_M) \rightarrow \text{coker}(\hat{\lambda}_M).$$

We have the following commutative diagram with exact rows [CP23, Remark 3.3].

$$(4.1) \quad \begin{array}{ccccccc} & & & \overline{H_2(M; \Lambda)^*} & \longrightarrow & \text{coker}(\hat{\lambda}_M) & \longrightarrow 0 \\ & \nearrow \hat{\lambda}_M & & \uparrow \text{evoPD} & & \downarrow D_M & \\ 0 & \longrightarrow & H_2(M; \Lambda) & & & & \\ & \searrow j & & H_2(M, \partial M; \Lambda) & \longrightarrow & H_1(\partial M; \Lambda) & \longrightarrow 0 \end{array}$$

The lower sequence arises from the long exact sequence of the pair $(M, \partial M)$. The middle vertical arrow is evaluation after Poincaré-Lefschetz duality with twisted coefficients. The diagram therefore induces an isomorphism of Λ -modules $D_M : \text{coker}(\hat{\lambda}_M) \rightarrow H_1(\partial M; \Lambda)$.

Definition 4.5 ([CP23, Definition 2.6]). The *boundary* of an isometry $\alpha \in \text{Isom}(\lambda_M)$ is

$$\Pi_M(\alpha) = D_M \circ [\alpha^{-*}] \circ D_M^{-1} : H_1(\partial M; \Lambda) \rightarrow H_1(\partial M; \Lambda)$$

This defines a homomorphism

$$\begin{aligned} \Pi_M : \text{Isom}(\lambda_M) &\rightarrow \text{Aut}_\Lambda(H_1(\partial M; \Lambda)) \\ \alpha &\rightarrow \Pi_M([\alpha^{-*}]). \end{aligned}$$

Denote the image of Π_M by $\Gamma_M \leq \text{Aut}(H_1(\partial M; \Lambda))$.

Theorem 4.6. Let M be a compact 4-manifold with $\pi_1 M \cong \mathbb{Z}$ whose boundary is ribbon and has $H_1(\partial M; \Lambda)$ a torsion Λ -module. Fix a homeomorphism $h : \partial M \rightarrow \partial M$ which intertwines the inclusion map $\pi_1 \partial M \rightarrow \pi_1 M \cong \mathbb{Z}$. Assume that a lift h_* of h to $H_1(\partial M; \Lambda)$ is the boundary (Definition 4.5) of an isometry in $\text{Isom}(\lambda_M)$, that is $h_* \in \Gamma_M$. Then there exists an orientation preserving homeomorphism $H : M \rightarrow M$ with $H|_{\partial M} = h$.

Proof. This is a special case of Theorem 4.1 in [CP23]. $\square$

4.3. The image of Π_M and the Blanchfield pairing. The group Γ_M is governed by additional structure on $H_1(\partial M; \Lambda)$, namely the *Blanchfield pairing* on ∂M . Let $\mathcal{Q}$ denote the field of fractions of Λ . As the boundary of M is ribbon, inclusion induces a surjection $\pi_1(\partial M) \rightarrow \mathbb{Z}$. Moreover, we've assumed that $H_1(\partial M; \Lambda)$ is a torsion Λ -module. It follows that $H_1(\partial M; \Lambda)$ is equipped with a sesquilinear, Hermitian, non-singular *linking form*

$$\text{Bl}_{\partial M} : H_1(\partial M; \Lambda) \times H_1(\partial M; \Lambda) \rightarrow \mathcal{Q}/\Lambda$$

referred to as the Blanchfield pairing on ∂M (see [CP23, Definition 3.4] for a precise definition). Let $\text{Isom}(\text{Bl}_{\partial M})$ denote the isometry group of $\text{Bl}_{\partial M}$.

Proposition 4.7. The boundary of an isometry of λ_M is an isometry of the Blanchfield pairing $\text{Bl}_{\partial M}$, that is, $\Gamma_M \leq \text{Isom}(\text{Bl}_{\partial M})$.

Proof. This is [CP23, Propostion 3.5] in combination with [CP23, Lemma 2.5]. $\square$

Recall that a surface Σ in a 4-manifold X is a $\mathbb{Z}$ -surface if $\pi_1(X - \Sigma) \cong \mathbb{Z}$. Conway-Powell identify the group $\text{Isom}(\text{Bl}_{\partial M})$ explicitly when M is a $\mathbb{Z}$ -surface exterior (this identification will be used in Section 5). To state the results, we first describe the boundary of a $\mathbb{Z}$ -surface exterior (in the rel boundary setting). If $\Sigma \subset D^4$ is a properly embedded $\mathbb{Z}$ -surface with boundary a knot $K \subset S^3$, let $D_\Sigma^4 = D^4 - \nu\Sigma$. Then $\partial D_\Sigma^4 \cong \Sigma \times S^1 \cup_\partial E_K$ where E_K denotes the knot exterior of K in S^3 . Mayer-Vietoris with Λ -coefficients shows that

$$(4.2) \quad H_1(\partial D_\Sigma^4; \Lambda) \cong H_1(\Sigma \times S^1; \Lambda) \oplus H_1(E_K; \Lambda).$$

Moreover, $H_1(\Sigma \times S^1; \Lambda) \cong (\Lambda/(t-1))^{\oplus 2g}$ where g is the genus of Σ .

Proposition 4.8. *For a properly embedded $\mathbb{Z}$ -surface $\Sigma \subset D^4$ with boundary a knot $K \subset S^3$, the isomorphism (4.2) induces a decomposition*

$$\text{Isom}(\text{Bl}_{\partial D_\Sigma^4}) \cong \text{Isom}(\text{Bl}_{\Sigma \times S^1}) \oplus \text{Isom}(\text{Bl}_{E_K})$$

and $\text{Isom}(\text{Bl}_{\Sigma \times S^1}) \cong \text{Sp}(H_1(\Sigma; \mathbb{Z}))$.

Proof. This is [CP23, Propositions 5.6 and 5.7]. $\square$

4.4. Extending mapping classes. We now record an application of Theorem 4.6 in the language of extendable mapping classes. For a compact surface Σ , let

$$\Psi : \text{Mod}(\Sigma) \rightarrow \text{Sp}(H_1(\Sigma; \mathbb{Z}))$$

be the *symplectic representation* of $\text{Mod}(\Sigma)$. The kernel of Ψ is the *Torelli subgroup* of $\text{Mod}(\Sigma)$, denoted $\mathcal{I}(\Sigma)$.

Corollary 4.9. *Let $\Sigma \subset D^4$ be a properly embedded $\mathbb{Z}$ -surface with a single boundary component. Then*

$$\mathcal{I}(\Sigma) \leq \mathcal{E}^{\text{TOP}}(D^4, \Sigma).$$

Proof. This follows from Theorem 4.6 and the methods in the proof of [CP23, Theorem 5.8]. Fix a homeomorphism $f : \Sigma \rightarrow \Sigma$ which is the identity on $\partial\Sigma$ and acts trivially on $H_1(\Sigma; \mathbb{Z})$. Write $\partial\Sigma = K \subset \partial D^4$ and E_K for the exterior of the knot K . Let $D_\Sigma^4 = D^4 - \nu\Sigma$ be the surface exterior. The boundary of D_Σ^4 is obtained from the knot exterior E_K by gluing in $\Sigma \times S^1$ along ∂E_K . Define a homeomorphism $h : \partial D_\Sigma^4 \rightarrow \partial D_\Sigma^4$ by

$$h = (f \times \text{Id}_{S^1}) \cup_\partial \text{Id}_{E_K}.$$

The homeomorphism h intertwines $\pi_1 \partial D_\Sigma^4 \rightarrow \mathbb{Z}$ as it is the identity on ∂E_K . In particular, we may consider lifts to automorphisms of $H_1(\partial D_\Sigma^4; \Lambda)$. Recall that Mayer-Vietoris with Λ -coefficients yields an isomorphism [CP23, Lemma 5.5]

$$H_1(\partial D_\Sigma^4; \Lambda) \cong H_1(\Sigma \times S^1; \Lambda) \oplus H_1(E_K; \Lambda).$$

In particular, $h_* : H_1(\partial D_\Sigma^4; \Lambda) \rightarrow H_1(\partial D_\Sigma^4; \Lambda)$ is the identity. By Theorem 4.6, there exists a homeomorphism $H : D_\Sigma^4 \rightarrow D_\Sigma^4$ extending h . Extend H over ∂D^4 by the identity. It follows that $[f] \in \mathcal{E}_\partial^{\text{TOP}}(D^4, \Sigma)$. $\square$

Remark 4.10. By the Alexander trick, the homeomorphism $H : (D^4, \Sigma) \rightarrow (D^4, \Sigma)$ constructed in the proof of Corollary 4.9 is topologically isotopic to the identity relative to the boundary ∂D^4 .

When the equivariant intersection form can be computed explicitly, isometries may be constructed directly. We carry this out for pushed in Seifert surfaces in the 4-ball.

4.5. Pushed in Seifert surfaces in D^4 . Let $F' \subset S^3$ be a Seifert surface for a knot K . Let $\iota: F' \times [0, 1] \rightarrow D^4$ be an isotopy which pushes the interior of F' into the interior of D^4 . Our pushed in Seifert surface is $F = \iota(F' \times \{1\})$. Let $V_F = D^4 - \nu F$ denote the exterior of F .

Proposition 4.11. *The pushed in Seifert surface $F \subset D^4$ is a $\mathbb{Z}$ -surface, that is, $\pi_1(V_F) \cong \mathbb{Z}$.*

Proof. By [GS99, Proposition 6.21], the exterior V_F admits a handle decomposition with only a single 1-handle (recall that F is, by definition, connected). Therefore $\pi_1(D^4 - \nu F)$ is a quotient of $\mathbb{Z}$. The normal bundle νF to F in D^4 is the trivial disk bundle $F \times D^2$ over F (it is an orientable disk bundle over F , and F is homotopy equivalent to a wedge of circles). The long exact sequence of the pair $(D^4 - F, \partial \nu F) \simeq (D^4 - F, \nu F - F)$, and excision of F applied to the pair $(D^4, \nu F)$, shows that $H_1(V_F; \mathbb{Z}) \cong \mathbb{Z}$. $\square$

Let $\widetilde{V}_F$ denote the universal cover of V_F . We begin by describing the standard model for $\widetilde{V}_F$. For details, we refer the reader to [Ko89, Page 537], [COT04, proof of Lemma 5.4] as well as [AK80]. Note that this procedure works for any pushed in F^{n-2} in an n -ball D^n .

Cut D^4 along the trace $S = \iota(F' \times [0, 1])$ of the isotopy ι which pushed the interior of F' into the interior of D^4 . The result $D^4 - \nu S$ is homeomorphic to D^4 (c.f. [GS99, Proposition 6.21]). This identification provides a thickening $\bar{F}' \subset \partial D^4$ of the interior of the Seifert surface F' in the boundary of D^4 . Precisely, there is an embedding of

$$\bar{F}' = F' \times [-1, 1] / (x, t) \sim (x, t'), \forall x \in \partial F', t, t' \in [-1, 1]$$

in $\partial(D^4 - \nu S) \cong \partial D^4$. Take countably many copies $t^k D^4$, $k \in \mathbb{Z}$ of D^4 . Fix $\epsilon > 0$ small. Set

$$\bar{F}'_+(k) = \{(x, t) \in \bar{F}' \mid x \in \text{Int}(F'), t \in [\epsilon, 1]\} \subset t^k D^4,$$

and

$$\bar{F}'_-(k) = \{(x, t) \in \bar{F}' \mid x \in \text{Int}(F'), t \in [-1, -\epsilon]\} \subset t^k D^4.$$

Let

$$\widetilde{V}_F = \bigsqcup_{k \in \mathbb{Z}} t^k D^4 / \sim$$

where we identify $\bar{F}'_+(k)$ with $\bar{F}'_-(k-1)$ via the map $h: \bar{F}'_+(k) \rightarrow \bar{F}'_-(k-1)$ that sends (x, t) to $(x, -t)$.

The connecting homomorphism

$$\delta: H_2(\widetilde{V}_F; \mathbb{Z}) \rightarrow H_1(F'; \mathbb{Z}) \otimes_{\mathbb{Z}} \Lambda$$

for the Mayer-Vietoris sequence of the decomposition $A = \cup_{k \text{ even}} t^k D^4$, $B = \cup_{k \text{ odd}} t^k D^4$, $A \cap B = \cup_{k \in \mathbb{Z}} \bar{F}'_+(k)$ gives an isomorphism of Λ -modules [Ko89, Page 538]. Therefore $H_2(\widetilde{V}_F; \mathbb{Z})$ is a free Λ -module.

4.6. Basing $H_2(V_F; \Lambda)$ and computing λ_{V_F} . To obtain a Λ -basis for $H_2(V_F; \Lambda)$, fix a collection $\{a_i\}_{i=1}^{2g}$ of oriented simple closed curves on F' that give a $\mathbb{Z}$ -basis for $H_1(F'; \mathbb{Z})$. Take immersed 2-disks $(\Sigma_i^+, \partial \Sigma_i^+) \subset (tD^4, \bar{F}'_+(1))$ and $(\Sigma_i^-, \partial \Sigma_i^-) \subset (t^0 D^4, \bar{F}'_-(0))$ such that $\partial \Sigma_i^+ = (a_i, \epsilon) \in \bar{F}'_+(1)$ and $\partial \Sigma_i^- = (-a_i, -\epsilon)$; we identify (a_i, ϵ) with a_i^+ , a pushoff of a_i in the positive normal direction to F' in S^3 . Similarly, we identify $(-a_i, -\epsilon)$ with $-a_i^-$, which denotes a_i with opposite orientation pushed off in the negative normal direction to F' . Set $\Sigma_i = \Sigma_i^+ \cup \Sigma_i^-$. Then the spheres $\{\Sigma_i\}_{i=1}^{2g}$ represent a Λ -basis for $H_2(\widetilde{V}_F; \mathbb{Z})$. Let S denote the Seifert form of F' relative to the basis $\{a_i\}_{i=1}^{2g}$ for $H_1(F'; \mathbb{Z})$. In the constructed basis, the equivariant intersection form λ_{V_F} is a deformation of S .

Proposition 4.12. *The spheres $\Sigma_1, \dots, \Sigma_{2g}$ constructed by doubling along cycles above give a basis $\mathcal{C}$ for the free Λ -module $H_2(\widetilde{V}_F; \mathbb{Z})$. In the basis $\mathcal{C}$ the equivariant intersection form*

$$\begin{aligned} \lambda_{V_F}: H_2(\widetilde{V}_F; \mathbb{Z}) \times H_2(\widetilde{V}_F; \mathbb{Z}) &\rightarrow \Lambda \\ (\Sigma_j, \Sigma_i) &\mapsto \sum_k (\Sigma_i \cdot t^k \Sigma_j) t^{-k}. \end{aligned}$$

is given by the matrix $Q \in M_{2g}(\Lambda)$ where

$$Q_{ij} = \lambda_{V_F}(\Sigma_i, \Sigma_j) = (1-t)S_{ji} + (1-t^{-1})S_{ij}.$$

Proof. A reference is [Ko89]. See also e.g. the proof of [COT04, Lemma 5.4] (note our conventions for λ_{V_F}). $\square$

Henceforth, we identify $F \subset D^4$ and $F' \subset S^3$. Let $\delta : H_2(V_F, \partial V_F; \Lambda) \rightarrow H_1(\partial V_F; \Lambda)$ denote the connecting homomorphism in the long exact sequence of the pair $(V_F, \partial V_F)$ with Λ coefficients. Recall that $H_1(\partial V_F; \Lambda) \cong H_1(F \times S^1; \Lambda) \oplus H_1(E_K; \Lambda)$ (4.2) and $H_1(F \times S^1; \Lambda) = H_1(F \times \mathbb{R}; \mathbb{Z}) \cong H_1(F; \mathbb{Z})$.

Lemma 4.13. *Fix $i \in \{1, \dots, 2g\}$. Then $\delta(\Sigma_i^+) = [a_i] \in H_1(F; \mathbb{Z}) \subset H_1(\partial V_F; \Lambda)$.*

Proof. The connecting homomorphism is

$$\delta : H_2(\widetilde{V_F}, \partial \widetilde{V_F}; \mathbb{Z}) \rightarrow H_1(\partial \widetilde{V_F}; \mathbb{Z}).$$

Set $\bar{F}'(k) = \{(x, t) \in \bar{F}' \mid t \in [-\epsilon, \epsilon]\} \subset t^k D^4$. Then

$$\partial \widetilde{V_F} = \bigcup_{k \in \mathbb{Z}} \bar{F}'(k) \cup \widetilde{E_K},$$

where $\bigcup_{k \in \mathbb{Z}} \bar{F}'(k) \cong F \times \mathbb{R}$ and $\widetilde{E_K}$ denotes the universal abelian cover of E_K . Here we implicitly use the standard description of $\widetilde{E_K}$ (c.f. [Rol76]). The decomposition (4.2) is $H_1(\partial \widetilde{V_F}; \mathbb{Z}) \cong H_1(F \times \mathbb{R}; \mathbb{Z}) \oplus H_1(\widetilde{E_K}; \mathbb{Z})$. Now $\delta(\Sigma_i^+) = [\partial \Sigma_i^+] = [(a_i, \epsilon)] \in H_1(F \times \mathbb{R}; \mathbb{Z})$. $\square$

4.7. Computing the boundary homomorphism Π_{V_F} . We now compute the restriction of the boundary of an isometry of λ_{V_F} to $H_1(F \times S^1; \Lambda)$. Precisely, for $\alpha \in \text{Isom}(\lambda_{V_F})$ and

$$\Pi_{V_F} : \text{Isom}(\lambda_{V_F}) \rightarrow \text{Isom}(\text{Bl}_{\partial V_F})$$

we compute $\Pi_{V_F}(\alpha)|_{H_1(F \times S^1; \Lambda)}$ (see Proposition 4.8).

Let $H^* = \overline{H_2(V_F; \Lambda)}^*$ (the dual of $H_2(V_F; \Lambda)$ but with Λ -module structure $p \cdot x = \bar{p}x$). Write $\lambda = \lambda_{V_F}$. Recall that we have an identification $D_{V_F} : \text{coker}(\widehat{\lambda}_{V_F}) \rightarrow H_1(\partial V_F; \Lambda)$ (Definition 4.5), and that the map

$$H_2(V_F, \partial V_F; \Lambda) \xrightarrow{\text{PD}^{-1}} H^2(V_F; \Lambda) \xrightarrow{\text{ev}} H^*$$

is identified with the currying map of the equivariant intersection form

$$\lambda^\delta : H_2(V_F; \Lambda) \times H_2(V_F, \partial V_F; \Lambda) \rightarrow \Lambda.$$

A basis for the $H_1(F \times S^1; \Lambda)$ factor in $\text{coker}(\widehat{\lambda}_{V_F})$ is given by $[\lambda^\delta(-, \Sigma_i^+)] \in \text{coker}(\widehat{\lambda}_{V_F})$ for $i = 1, \dots, 2g$. The Λ -basis $\{\Sigma_i\}_{i=1}^{2g}$ for $H_2(V_F; \Lambda)$ gives the dual basis $\{\Sigma_i^*\}_{i=1}^{2g}$ for H^* . Recall that S denotes the Seifert form of F in the basis $\{a_i\}_{i=1}^{2g}$ for $H_1(F; \mathbb{Z}) \cong H_1(F \times S^1; \Lambda)$, and that $j : H_2(V_F; \Lambda) \rightarrow H_2(V_F, \partial V_F)$ is the homomorphism induced by inclusion of pairs (4.1).

Lemma 4.14. *We have*

$$\lambda^\delta(-, \Sigma_i^+) = \sum_{j=1}^{2g} (S_{ij} - S_{ji}t^{-1})\Sigma_j^*,$$

and

$$j(\Sigma_i) = (1 - t^{-1})\Sigma_i^+.$$

Therefore $(1 - t)\lambda^\delta(-, \Sigma_i^+) = \lambda(-, \Sigma_i) = 0 \in \text{coker}(\widehat{\lambda}_{V_F})$. Let $\alpha \in \text{Isom}(\lambda_{V_F})$ and let A denote the matrix of α with respect to the basis $\mathcal{C}$ (Proposition 4.12). Consider the $\mathbb{Z}$ -submodule $U \subset \text{coker}(\widehat{\lambda}_{V_F})$ spanned by the $\mathbb{Z}$ -basis $\{\lambda^\delta(-, \Sigma_i^+)\}_{i=1}^{2g} = \mathcal{B}$. Then the matrix representing the action of α^* on U is given by $A^{-1} \pmod{t-1}$ in the basis $\mathcal{B}$.

Proof. We have $\lambda^\delta(-, \Sigma_i^+) = \sum_{j=1}^{2g} a_j \Sigma_j^*$ where $a_j = \lambda^\delta(\Sigma_j, \Sigma_i^+) = \sum_{k \in \mathbb{Z}} (\Sigma_i^+ \cdot t^k \Sigma_j) t^{-k}$. We have $\Sigma_i^+ \cdot t^k \Sigma_j = 0$ for $k \neq 0, 1$. For $k = 0$, we have $\Sigma_i^+ \cdot \Sigma_j = \Sigma_i^+ \cdot \Sigma_j^+ = lk(a_i^+, a_j^+) = lk(a_i, a_j^+) = S_{ij}$. For $k = 1$ we have $\Sigma_i^+ \cdot t \Sigma_j = \Sigma_i^+ \cdot t \Sigma_j^- = lk(a_i^+, -a_j^-) = -S_{ji}$. This shows the first claim.

For the second claim, we have a commutative diagram

$$\begin{array}{ccc} H_2(V_F; \Lambda) & \xrightarrow{\hat{\lambda}} & H^* \\ j \downarrow & \nearrow \hat{\lambda}^\delta & \\ H_2(V_F, \partial V_F; \Lambda) & & \end{array}$$

where j comes from the exact sequence

$$0 \rightarrow H_2(V_F; \Lambda) \xrightarrow{j} H_2(V_F, \partial V_F; \Lambda) \xrightarrow{\delta} H_1(\partial V_F; \Lambda) \rightarrow 0.$$

The basis $\{\Sigma_i\}$ for $H_2(V_F; \Lambda)$ has the form $\Sigma_i = \Sigma_i^+ \cup \Sigma_i^-$ and $j(\Sigma_i) = \Sigma_i^+ + \Sigma_i^-$. We have (using Proposition 4.12 which computes $\lambda(\Sigma_j, \Sigma_i)$),

$$\begin{aligned} (1-t)S_{ij} + (1-t^{-1})S_{ji} &= \lambda(\Sigma_j, \Sigma_i) \\ &= \lambda^\delta(\Sigma_j, j(\Sigma_i)) \\ &= \lambda^\delta(\Sigma_j, \Sigma_i^+ + \Sigma_i^-) \\ &= S_{ij} - S_{ji}t^{-1} + \lambda^\delta(\Sigma_j, \Sigma_i^-). \end{aligned}$$

Hence we have

$$\lambda^\delta(\Sigma_j, \Sigma_i^-) = S_{ji} - tS_{ij},$$

which implies that

$$\lambda^\delta(-, \Sigma_i^-) = \sum_{j=1}^{2g} \lambda^\delta(\Sigma_j, \Sigma_i^-) \Sigma_j^* = \sum_{j=1}^{2g} (S_{ji} - tS_{ij}) \Sigma_j^* = -t\lambda^\delta(-, \Sigma_i^+).$$

Since $\hat{\lambda}^\delta$ is an isomorphism, we get that $j(\Sigma_i) = \Sigma_i^+ + \Sigma_i^- = (1-t^{-1})\Sigma_i^+$.

For the last claim, write $\alpha^{-1}(\Sigma_i) = \sum_j b_{ji} \Sigma_j$, then $\alpha^* \lambda^\delta(-, \Sigma_i^+) = \lambda^\delta(\alpha(-), \Sigma_i^+)$. We have

$$\begin{aligned} (1-t)\lambda^\delta(\alpha(-), \Sigma_i^+) &= \lambda^\delta(\alpha(-), j(\Sigma_i)) \\ &= \lambda(\alpha(-), \Sigma_i) \\ &= \lambda(-, \alpha^{-1}(\Sigma_i)) \\ &= \sum_j \bar{b}_{ji} \lambda(-, \Sigma_j) \\ &= (1-t) \sum_j \bar{b}_{ji} \lambda^\delta(-, \Sigma_j^+). \end{aligned}$$

Since Λ is an integral domain, we obtain

$$\alpha^* \lambda^\delta(-, \Sigma_i^+) = \sum_j \bar{b}_{ji} \lambda^\delta(-, \Sigma_j^+).$$

Note that $H^* = \text{Hom}_\Lambda(H_2(V_F; \Lambda), \Lambda)$ is equipped with the Λ -module structure $p \cdot h = \bar{p}h$ so that the maps $\hat{\lambda}, \hat{\lambda}^\delta$ become Λ -module maps, so the final claim follows. $\square$

Relative to our basis $\mathcal{C}$ for $H_2(V_F; \Lambda)$ we have

$$\text{Isom}(H_2(V_F; \Lambda), \lambda) = \{A \in GL_{2g}(\Lambda) \mid A^T Q \bar{A} = Q\}.$$

We have $Q = (1-t)P$, where $P = S^T - t^{-1}S$. Any matrix $A \in GL_{2g}(\Lambda)$ with $A^T Q \bar{A} = Q$ must satisfy

$$(4.3) \quad A^T P \bar{A} = P.$$

Note that $P \pmod{t-1} = S^T - S$ is the intersection form on $H_1(F; \mathbb{Z})$ in our basis of curves $\{a_i\}_{i=1}^{2g}$ [Rol76, Chapter 8, Exercise 7]. Therefore $A \pmod{t-1}$ has image in $\text{Sp}_{2g}(\mathbb{Z})$.

Corollary 4.15. *The map*

$$\begin{aligned} \Pi_{V_F} : \text{Isom}(\lambda_{V_F}) &\rightarrow \text{Isom}(\text{Bl}_{V_F}) \\ \alpha &\mapsto [\alpha^{-*}] \end{aligned}$$

of Definition 4.5 is given by $A \mapsto A \pmod{t-1}$ on the $H_1(F \times S^1; \Lambda)$ factor of $H_1(\partial V_F; \Lambda)$ where A is the matrix of α with respect to the basis $\mathcal{C}$ of $H_2(V_F; \Lambda)$ considered in Lemma 4.14.

Proof. This follows from Lemma 4.14 and the definition of Π_{V_F} . $\square$

4.8. Extending mapping classes with explicit transvections. We now use the computation of λ_{V_F} (Proposition 4.12) to write down isometries of λ_{V_F} . We then use these isometries to exhibit certain Dehn twists as topologically extendable mapping classes.

Definition 4.16. Let H be a free Λ -module equipped with a sesquilinear Hermitian pairing $\lambda : H \times H \rightarrow \Lambda$. We say λ is divisible by $r \in \Lambda$ if for any $x, y \in H$, there exists some $s \in \Lambda$ (depending on x and y) such that $\lambda(x, y) = rs$.

Remark 4.17. By Proposition 4.12, λ_{V_F} is divisible by $1 - t$. Indeed, in the basis $\mathcal{C} = \{\Sigma_i\}_{i=1}^{2g}$, we have $\lambda = (1 - t)P$ where $P = S^T - t^{-1}S$.

Proposition 4.18. *Let H be a free Λ -module equipped with a sesquilinear Hermitian pairing $\lambda : H \times H \rightarrow \Lambda$. Fix $v \in H$, $p \in \Lambda$. Consider the transvection $T_{v,p} : H \rightarrow H$ defined by*

$$T_{v,p}(x) = x + p\lambda(x, v)v.$$

We have the following properties.

- (a) *If $p + \bar{p} + p\bar{p}\lambda(v, v) = 0$, then $\lambda(T_{v,p}(x), T_{v,p}(y)) = \lambda(x, y)$ for all $x, y \in H$. If $\lambda(v, v) = 0$ and $p + \bar{p} = 0$, then $T_{v,p} \in \text{Isom}(H, \lambda)$, with inverse $T_{v,-p}$.*
- (b) *If λ is divisible by $1 - t$, we may allow $p = s/(1 - t)$ with $s \in \Lambda$ in the definition of $T_{v,p}$ and in the item above (part (a)).*
- (c) *Assume that λ is divisible by $(1 - t)$ and $\lambda(v, v) = \pm(1 - t)(1 - t^{-1})$. Let $p = \pm t/(1 - t)$. Then $p + \bar{p} + p\bar{p}\lambda(v, v) = 0$ and $T_{v, \frac{\pm 1}{1-t}}$ is the inverse of $T_{v,p}$. Therefore $T_{v,p} \in \text{Isom}(H, \lambda)$.*

Proposition 4.19. *Let $F \subset D^4$ be a pushed in Seifert surface with exterior $V_F = D^4 - \nu F$. Fix a simple closed curve $a \subset F$. Assume that $\text{lk}(a, a^+) = 0$ or ± 1 . Let $\Sigma_a \subset \widetilde{V}_F$ denote a 2-cycle obtained by doubling along push-offs of $a \subset F$ as in Proposition 4.12. Let*

$$p_a = \begin{cases} \frac{t+1}{1-t} & \text{if } \text{lk}(a, a^+) = 0, \\ \frac{\pm t}{1-t} & \text{if } \text{lk}(a, a^+) = \pm 1. \end{cases}$$

Then $T_{\Sigma_a, p_a} \in \text{Isom}(\lambda_{V_F})$. Moreover, the restriction $\Pi_{V_F}(T_{\Sigma_a, p_a})|_{H_1(F \times S^1; \Lambda)}$ is the ordinary transvection $T_{[a]}^{\text{lk}(a, a^+)} \in \text{Sp}(H_1(F; \mathbb{Z}))$ along $[a] \in H_1(F; \mathbb{Z})$ if $\text{lk}(a, a^+) = \pm 1$ and the square $T_{[a]}^2$ if $\text{lk}(a, a^+) = 0$. On the other hand, $\Pi_{V_F}(T_{\Sigma_a, p_a})|_{H_1(E_K; \Lambda)}$ is the identity in either case.

Proof. That $T_{\Sigma_a, p_a} \in \text{Isom}(\lambda_{V_F})$ follows from Proposition 4.18. Indeed, if $\text{lk}(a, a^+) = 0$, then $\lambda_{V_F}(\Sigma_a, \Sigma_a) = 0$ and $p_a + \bar{p}_a = 0$. If $\text{lk}(a, a^+) = \pm 1$, then $\lambda(\Sigma_a, \Sigma_a) = \pm(1 - t)(1 - t^{-1})$ and the claim follows from part (c) of Proposition 4.18.

Recall that $U \subset \text{coker}(\widehat{\lambda}_{V_F})$ is the $\mathbb{Z}$ -submodule spanned by $\lambda^\delta(-, \Sigma_i^+)$ for $i = 1, \dots, 2g$. Under the isomorphism

$$D_{V_F} : \text{coker}(\widehat{\lambda}_{V_F}) \rightarrow H_1(\partial V_F; \Lambda)$$

the submodule U corresponds to the $H_1(F \times S^1; \Lambda) \cong H_1(F; \mathbb{Z})$ factor. By definition of Π_{V_F} , we must compute the action T_{Σ_a, p_a}^{-*} on U . In the notation of Lemma 4.14, we have, first working on

the level of the dual module H^* ,

$$\begin{aligned}
(1-t)T_{\Sigma_a, p_a}^{-*}(\lambda^\delta(-, \Sigma_i^+)) &= \lambda(T_{\Sigma_a, p_a}^{-1}(-), \Sigma_i) \\
&= \lambda(-, T_{\Sigma_a, p_a}(\Sigma_i)) \\
&= \lambda(-, \Sigma_i + p_a \lambda(\Sigma_i, \Sigma_a) \Sigma_a) \\
&= \lambda(-, \Sigma_i) + \overline{p_a \lambda(\Sigma_i, \Sigma_a)} \lambda(-, \Sigma_a) \\
&= (1-t)(\lambda^\delta(-, \Sigma_i^+) + \overline{p_a \lambda(\Sigma_i, \Sigma_a)} \lambda^\delta(-, \Sigma_a^+)),
\end{aligned}$$

and hence

$$T_{\Sigma_a, p_a}^{-*}(\lambda^\delta(-, \Sigma_i^+)) = \lambda^\delta(-, \Sigma_i^+) + \overline{p_a \lambda(\Sigma_i, \Sigma_a)} \lambda^\delta(-, \Sigma_a^+).$$

The connecting map $\delta: H_2(V_F, \partial V_F; \Lambda) \rightarrow H_1(\partial V_F; \Lambda)$ sends Σ_i^+ to $[a_i]$, and Σ_a^+ to $[a]$ (Lemma 4.13). It follows that

$$\Pi_{V_F}(T_{\Sigma_a, p_a})[a_i] = [a_i] + \overline{p_a \lambda(\Sigma_i, \Sigma_a)}[a].$$

By (the proof of) Proposition 4.12,

$$\lambda(\Sigma_i, \Sigma_a) = (1-t)(\text{lk}(a, a_i^+) - t^{-1}\text{lk}(a_i, a^+)).$$

Therefore

$$p_a \lambda(\Sigma_i, \Sigma_a) = \begin{cases} (t+1)(\text{lk}(a, a_i^+) - t^{-1}\text{lk}(a_i, a^+)) & \text{if } \text{lk}(a, a^+) = 0, \\ \pm t(\text{lk}(a, a_i^+) - t^{-1}\text{lk}(a_i, a^+)) & \text{if } \text{lk}(a, a^+) = \pm 1. \end{cases}$$

Recall that $t = 1$ in the $H_1(F \times S^1; \Lambda)$ factor, and that, for the Seifert form S , the difference $S^T - S$ is the oriented intersection matrix of the curves $\{a_i\}_{i=1}^{2g}$ [Rol76, Chapter 8, Exercise 7]. Therefore

$$\text{lk}(a, a_i^+) - \text{lk}(a_i, a^+) = \langle a_i, a \rangle$$

where $\langle -, - \rangle$ denotes the oriented intersection pairing on $H_1(F; \mathbb{Z})$. It follows that

$$\Pi_{V_F}(T_{\Sigma_a, p_a})[a_i] = \begin{cases} [a_i] + 2\langle a_i, a \rangle[a] & \text{if } \text{lk}(a, a^+) = 0, \\ [a_i] \pm \langle a_i, a \rangle[a] & \text{if } \text{lk}(a, a^+) = \pm 1. \end{cases}$$

This is the desired (power of a) transvection in either case.

To see that $\Pi_{V_F}(T_{\Sigma_a, p_a})$ acts trivially on $H_1(E_K; \Lambda)$, use

$$\text{coker}(\widehat{\lambda}_{V_F}) \cong H_1(\partial V_F; \Lambda) \cong H_1(F \times S^1; \Lambda) \oplus H_1(E_K; \Lambda)$$

to exhibit $H_1(E_K; \Lambda)$ as the quotient $\text{coker}(\widehat{\lambda}_{V_F})/U$. By naturality, it suffices to show that T_{Σ_a, p_a}^* acts trivially on this quotient. For $\varphi \in H^*$,

$$T_{\Sigma_a, p_a}^*(\varphi) = \varphi + p_a \lambda(-, \Sigma_a) \varphi(\Sigma_a) = \varphi + p_a (1-t) \lambda^\delta(-, \Sigma_a^+) \varphi(\Sigma_a).$$

Now $[\lambda^\delta(-, \Sigma_a^+)] = 0$ in $\text{coker}(\widehat{\lambda}_{V_F})/U$, so $[T_{\Sigma_a, p_a}^*(\varphi)] = [\varphi] \in \text{coker}(\widehat{\lambda}_{V_F})/U$. □

We now generalize the Hopf band trick (Proposition 2.7) and the *tube isotopy*. The latter realizes the square of a Dehn twist along a curve bounding a zero-framed disk in the surface exterior as extendable (see [Hir05, Figure 1]). Note that both of these techniques require the presence of an embedded disk in the surface exterior.

Corollary 4.20. *Let $F \subset D^4$ be a pushed in Seifert surface for a knot. If $\gamma \subset F$ is a simple closed curve with $\text{lk}(\gamma, \gamma^+) = \pm 1$, then $T_\gamma \in \mathcal{E}_\partial^{\text{TOP}}(D^4, F)$. If $\gamma \subset F$ is a simple closed curve with $\text{lk}(\gamma, \gamma^+) = 0$, then $T_\gamma^2 \in \mathcal{E}_\partial^{\text{TOP}}(D^4, F)$.*

Proof. Proceed as in the proof of Corollary 4.9 using the explicit transvections constructed in Proposition 4.19. □

Proposition 4.20 does not address curves $\gamma \subset F$ with Seifert framing $|\text{lk}(\gamma, \gamma^+)| > 1$. In general, no claim can be made about such curves. Indeed, the spin-flexibility of unknots (Proposition 2.6) implies that there exist curves with any prescribed odd Seifert framing whose associated twists extend. On the other hand, we show in Lemma 5.6 that there exist curves with odd Seifert framing for which the associated twists do not extend. As Rochlin's form $q_F : H_1(F; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the mod 2 Seifert form, this is an obstruction to spin-flexibility.

5. OBSTRUCTING EXTENDABLE MAPPING CLASSES VIA THE EQUIVARIANT INTERSECTION FORM

In this section, we demonstrate how the equivariant intersection form on the surface exterior can be used to obstruct extendable mapping classes. We apply our obstruction to Milnor fibers of isolated hypersurface singularities of type ADE. In this setting, the image of the extendable subgroup under the mod 2 symplectic representation is precisely the associated mod 2 Weyl group.

5.1. Obstructing through the equivariant intersection form. We begin with a general obstruction to extending a mapping class of a pushed in Seifert surface in D^4 . We use notation introduced in Section 4. Let $F \subset D^4$ be a genus g Seifert surface for a knot $K \subset S^3$ with its interior pushed into the interior of D^4 . Let S be the Seifert matrix of F with respect to a fixed basis of curves $\{a_i\}_{i=1}^{2g} \subset F$ for $H_1(F; \mathbb{Z})$. Let $V_F = D^4 - \text{Int } \nu(F)$. Recall that the matrix Q of the equivariant intersection form on $H_2(V_F; \Lambda)$ in the Λ -basis $\mathcal{C}$ obtained from $\{a_i\}_{i=1}^{2g}$ (Proposition 4.12) is $Q = (1 - t)P$, where $P = S^T - t^{-1}S$. For any matrix A with entries in Λ , write $A(\pm 1)$ for the integer matrices $A \pmod{t \mp 1}$. The basis $\{a_i\}_{i=1}^{2g}$ determines a mod 2 symplectic representation $\Psi_2 : \text{Mod}(F) \rightarrow \text{Sp}_{2g}(\mathbb{Z}_2)$.

Lemma 5.1. *Let*

$$G_P = \{w \in \text{GL}_{2g}(\mathbb{Z}) \mid w^T P(-1)w = P(-1)\},$$

where $P(-1) = S^T + S$. If $f \in \mathcal{E}^{\text{TOP}}(D^4, F)$, then $\Psi_2(f)$ lies in the mod 2 reduction of G_P .

Proof. Suppose $f : D^4 \rightarrow D^4$ is an orientation-preserving homeomorphism fixing F setwise and ∂D^4 pointwise. Up to isotopy rel $\partial D^4 \cup F$, f restricts to a bundle map on a tubular neighbourhood $\nu(F)$ of F in D^4 , and the restriction $f|_{V_F}$ lifts to a homeomorphism $\tilde{f}$ of the universal cover $\tilde{V}_F$. This induces an isometry $\alpha_f = \tilde{f}_*$ of the equivariant intersection form $(H_2(V_F; \Lambda), Q)$. The restriction $f|_{\partial V_F} : \partial V_F \rightarrow \partial V_F$ intertwines the inclusion map $i : \pi_1(\partial V_F) \rightarrow \mathbb{Z}$. In particular, it induces an automorphism $(f|_{\partial V_F})_* : H_1(\partial V_F; \Lambda) \rightarrow H_1(\partial V_F; \Lambda)$. By (the proof of) [CP23, Proposition 3.11], the following diagram is commutative

$$\begin{array}{ccc} \text{coker}(\hat{\lambda}_{V_F}) & \xrightarrow{[\alpha_f^{-*}]} & \text{coker}(\hat{\lambda}_{V_F}) \\ D_{V_F} \downarrow & & \downarrow D_{V_F} \\ H_1(\partial V_F; \Lambda) & \xrightarrow{(f|_{\partial V_F})_*} & H_1(\partial V_F; \Lambda) \end{array}$$

Therefore $\Pi_{V_F}(\alpha_f) = (f|_{\partial V_F})_*$. By Proposition 4.7, $(f|_{\partial V_F})_* \in \text{Isom}(\text{Bl}_{\partial V_F})$. Recall that $\partial V_F = F \times S^1 \cup_{\partial} E_K$ and Mayer-Vietoris gives an isomorphism $H_1(\partial V_F; \Lambda) \cong H_1(E_K; \Lambda) \oplus H_1(F \times S^1; \Lambda)$. By Proposition 4.8,

$$\text{Isom}(\text{Bl}_{\partial V_F}) \cong \text{Isom}(\text{Bl}_{F \times S^1}) \oplus \text{Isom}(\text{Bl}_{E_K})$$

and $\text{Isom}(\text{Bl}_{F \times S^1}) \cong \text{Sp}(H_1(F; \mathbb{Z}))$. Write $(f|_{\partial V_F})_* = h_F \oplus h_K$ for $h_F \in \text{Isom}(\text{Bl}_{F \times S^1})$ and $h_K \in \text{Isom}(\text{Bl}_{E_K})$. Recall that the restriction of $\tilde{V}_F$ to the $F \times S^1$ part of ∂V_F is the covering $p : F \times \mathbb{R} \rightarrow F \times S^1$. It follows that the automorphism h_F fits into the following commutative diagram,

$$\begin{array}{ccccc} H_1(F \times \mathbb{R}; \mathbb{Z}) & \xrightarrow{p_*} & H_1(F \times S^1; \mathbb{Z}) & \xrightarrow{i_{F*}} & H_1(F \times D^2; \mathbb{Z}) \\ h_F \downarrow & & & & \downarrow (f|_{\nu F})_* \\ H_1(F \times \mathbb{R}; \mathbb{Z}) & \xrightarrow{p_*} & H_1(F \times S^1; \mathbb{Z}) & \xrightarrow{i_{F*}} & H_1(F \times D^2; \mathbb{Z}) \end{array}$$

where $i_F : F \times S^1 \rightarrow F \times D^2$ is inclusion into the closed tubular neighbourhood $\nu F \cong F \times D^2$. As $i_{F*} \circ p_*$ is an isomorphism, the induced action of $(f|_{\partial V_F})_*$ on the $H_1(F \times S^1; \Lambda)$ factor is $(f|_F)_*$.

On the other hand, we have computed the action of isometries of λ_{V_F} on the $H_1(F \times S^1; \Lambda)$ factor explicitly. In terms of our bases, the map

$$\Pi_{V_F} : \text{Isom}(H_2(V_F; \Lambda), Q) \rightarrow \text{Isom}(\text{Bl}_{\partial V_F})$$

is given by $A \mapsto A \pmod{t-1} =: A(1)$ on the $H_1(F \times S^1; \Lambda)$ factor (Corollary 4.15). It follows that if a mapping class $f : F \rightarrow F$ is extendable, its image in $\text{Sp}_{2g}(\mathbb{Z})$ determined by $f_* : H_1(F; \mathbb{Z}) \rightarrow H_1(F; \mathbb{Z})$ and $\{a_i\}_{i=1}^{2g}$ is the specialization of a matrix $A \in \text{Isom}(H_2(V_F; \Lambda), Q) \leq \text{GL}_{2g}(\Lambda)$ to $t = 1$.

The claim follows from specializing A to $t = -1$. Indeed, if $A \in \text{Isom}(H_2(V_F; \Lambda), Q)$, then $A^T P A = P$, so we have $A(-1)^T P(-1) A(-1) = P(-1)$, hence $A(-1) \in G_P$. Now $A(-1) \equiv A(1) \pmod{2}$ and $\Psi_2(f) = A(1) \pmod{2}$. $\square$

5.2. Application to Milnor fibers of ADE singularities. Let $f \in \mathbb{C}[x, y]$ be the germ of a simple isolated singularity at $0 \in \mathbb{C}^2$. Let $D^4(\epsilon)$ be a ball of radius ϵ centered at $0 \in \mathbb{C}^2$, and $D_\delta = \{z \in \mathbb{C} \mid |z| \leq \delta\}$. For $\epsilon > 0$ sufficiently small, the intersection $L = f^{-1}(0) \cap \partial D^4(\epsilon)$ is transverse and defines the *link* of the singularity [Mil68, Corollary 2.9]. Moreover, for $0 < \delta \ll \epsilon$, the restriction

$$f|_{f^{-1}(\partial D_\delta) \cap D^4(\epsilon)} : f^{-1}(\partial D_\delta) \cap D^4(\epsilon) \rightarrow \partial D_\delta$$

is a fiber bundle [Mil68]. The fiber of $f|_{f^{-1}(\partial D_\delta) \cap D^4(\epsilon)}$ above a point of ∂D_δ is the *Milnor fiber* Σ_f of f . The Milnor fiber of f is a compact, oriented surface with non-empty boundary.

Proposition 5.2. *The Milnor fiber of an isolated hypersurface singularity f is a $\mathbb{Z}$ -surface in a 4-ball. In fact, it is a pushed in Seifert surface.*

Proof. Milnor's fibration theorem [Mil68, Theorem 4.8] asserts that the link $L \subset \partial D^4(\epsilon)$ is fibered by the map

$$\frac{f}{|f|} : \partial D^4(\epsilon) - L \rightarrow S^1.$$

The fiber F_θ above $\theta \in S^1$ is the interior of a (minimal genus) Seifert surface for L [Mil68, Lemma 6.1]. The proof of [Mil68, Theorem 5.11] shows that $\Sigma_f = f^{-1}(c) \cap D^4(\epsilon)$ for $c \in \partial D_\delta$ is obtained by pushing F_θ (for $\theta = c/|c|$) into the interior of $D^4(\epsilon)$. Finally, pushed in Seifert surfaces have knot group $\mathbb{Z}$ (Proposition 4.11). $\square$

Simple isolated singularities are classified into ADE type, and are given by

$$(A_n) f(x, y) = x^2 + y^{n+1}, \quad (D_n) f(x, y) = xy^2 + x^{n-1},$$

and the exceptional types

$$(E_6) f(x, y) = x^3 + y^4, \quad (E_7) f(x, y) = x^3 + xy^3, \quad (E_8) f(x, y) = x^3 + y^5.$$

For each family, the following table describes the link of the singularity as the closure of a positive braid (see [Cas22, Example 2.5]).

Type	Positive Braid
A_n	σ_1^{n+1}
D_n	$\sigma_1^{n-2} \sigma_2 \sigma_1^2 \sigma_2$
E_6	$\sigma_1^3 \sigma_2 \sigma_1^3 \sigma_2$
E_7	$\sigma_1^4 \sigma_2 \sigma_1^3 \sigma_2$
E_8	$\sigma_1^5 \sigma_2 \sigma_1^3 \sigma_2$

Table 1. The links of singularities of type ADE as the closure of positive braids.

For each singularity $\Gamma = A_n, D_n, E_6, E_7$ or E_8 , the Milnor fiber F_Γ is, up to isotopy, the *standard Seifert surface* associated with the braid word (see [Rud83, Section 2] for a precise definition of the standard Seifert surface associated to a braid word). Depicted in Figure 1 is F_Γ for $\Gamma = A_8$. The Seifert surface F_Γ is equipped with a distinguished ordered configuration of curves $\mathcal{C}_\Gamma$ that give a basis for $H_1(F_\Gamma; \mathbb{Z})$ (c.f. [Fer25, Figure 1]) and have *intersection graph* (Definition 3.12) given by the corresponding ADE Dynkin diagram. We abuse notation and denote the Dynkin diagram associated to the singularity by Γ . Let S denote the Seifert matrix for F_Γ in the basis $\mathcal{C}_\Gamma$. Then

$P(-1) = S + S^T$ is the *Cartan matrix* associated to the Dynkin diagram (see [Baa13, Section 2]). Lemma 5.1 gives the following.

Theorem E. *Let Γ be a Coxeter–Dynkin diagram of type A_n with n even, E_6 or E_8 . Let $W_2(\Gamma)$ be the subgroup of $\mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ generated by the transvections in the homology classes of the curves in C_Γ . This is the mod 2 reduction of the Weyl group of type Γ . Then*

$$\Psi_2(\mathcal{E}^{\mathrm{TOP}}(D^4, F_\Gamma)) = W_2(\Gamma)$$

where $\Psi_2 : \mathrm{Mod}(F_\Gamma) \rightarrow \mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ is the mod 2 symplectic representation.

Proof. We begin with the containment $\Psi_2(\mathcal{E}^{\mathrm{TOP}}(D^4, F_\Gamma)) \leq W(\Gamma)$. By Lemma 5.1, $\Psi_2(\mathcal{E}^{\mathrm{TOP}}(D^4, F_\Gamma))$ is contained in the mod 2 reduction of

$$G_P = \{w \in \mathrm{GL}_{2g}(\mathbb{Z}) \mid w^T P(-1)w = P(-1)\},$$

where $P(-1) = S + S^T$ is the Cartan matrix associated to the Coxeter–Dynkin diagram Γ . The group G_P is the isometry group of the root lattice of type Γ . We now show the mod 2 reduction of G_P is precisely the mod 2 Weyl group $W_2(\Gamma)$. Recall that

$$G_P \cong W(\Gamma) \rtimes \mathrm{Aut}(\Gamma)$$

where $\mathrm{Aut}(\Gamma)$ denotes the automorphism group of the graph Γ and $W(\Gamma)$ the Weyl group of Γ . For Γ of type A_n or E_6 , we have $\mathrm{Aut}(\Gamma) = \langle \iota \rangle$, where ι is an involution; in these cases, for the longest element $\omega_0 \in W(\Gamma)$ with respect to the simple reflections determined by Γ , we have that $\iota = -\omega_0$ as an isometry of the lattice. For $\Gamma = E_8$, we have that $\mathrm{Aut}(\Gamma) = 1$. Hence the mod 2 reduction of the isometry group of the lattice agrees with the mod 2 reduction of the Weyl group $W(\Gamma)$.

We now prove that $W_2(\Gamma) \leq \Psi_2(\mathcal{E}^{\mathrm{TOP}}(D^4, F))$. The curves in C_Γ all bound disks in the exterior of $F_\Gamma \subset D^4$ with relative Euler number -1 (Remark 5.3). The Hopf band trick (Proposition 2.7) implies that

$$\mathcal{T}_{C_\Gamma} = \langle T_c : c \in C_\Gamma \rangle \leq \mathcal{E}^{\mathrm{DIFF}}(D^4, F) \leq \mathcal{E}^{\mathrm{TOP}}(D^4, F).$$

Therefore $W_2(\Gamma) \leq \Psi_2(\mathcal{E}^{\mathrm{TOP}}(D^4, F))$. □

Remark 5.3. The explicit construction of F_Γ and C_Γ exhibits the curves in C_Γ as unknots in an S^3 slice of a collar about ∂D^4 with Seifert framing -1 .

Let $q_{F_\Gamma} : H_1(F_\Gamma; \mathbb{Z}_2) \rightarrow \mathbb{Z}_2$ denote the mod 2 Seifert form on F_Γ (equivalently, Rochlin’s form). Symplectic automorphisms of $H_1(F_\Gamma; \mathbb{Z}_2)$ act on the set of quadratic forms on $H_1(F_\Gamma; \mathbb{Z}_2)$ by pullback.

Definition 5.4. Denote the stabilizer of q_{F_Γ} in $\mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ by $\mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))[q]$.

For any $c \in C_\Gamma$, we have $q_{F_\Gamma}(c) = 1$. Direct computation shows that if $q_{F_\Gamma}(c) = 1$, then $T_{[c]} \in \mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))[q_{F_\Gamma}]$. Let g be the genus of F_Γ . Consider reflections $\mathrm{Ref}_\alpha : \mathbb{Z}^{2g} \rightarrow \mathbb{Z}^{2g}$ given by

$$\mathrm{Ref}_\alpha(\beta) = \beta - \frac{2\beta \cdot \alpha}{\alpha \cdot \alpha} \alpha,$$

where $\beta \cdot \alpha$ denotes the bilinear form represented by $S^T + S$ in the basis C_Γ . Then $\mathrm{Ref}_c \equiv T_{[c]} \pmod{2}$. Since $W_2(\Gamma)$ is the subgroup generated by the $T_{[c]} : H_1(F_\Gamma; \mathbb{Z}_2) \rightarrow H_1(F_\Gamma; \mathbb{Z}_2)$ for $c \in C_\Gamma$, we obtain $W_2(\Gamma) \subset \mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))[q_{F_\Gamma}]$.

Remark 5.5. By order considerations, $W_2(\Gamma) \subset \mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))$ is a proper subgroup if $\Gamma = A_n$, E_6 or E_8 .

We now give an example of transvection in $\mathrm{Sp}(H_1(F_\Gamma; \mathbb{Z}_2))[q_{F_\Gamma}]$ that is not in $W_2(\Gamma)$ for $\Gamma = A_n$.

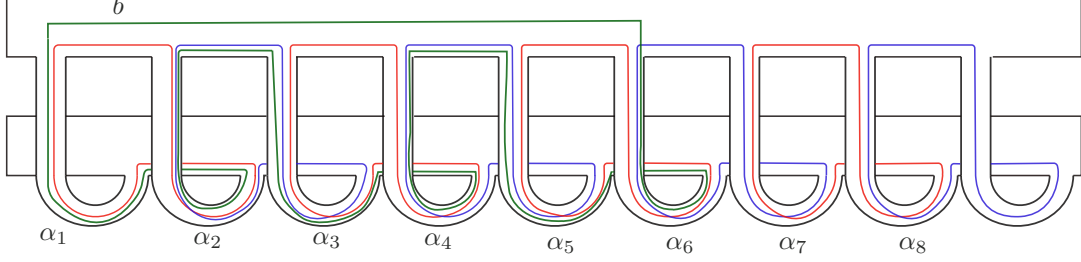


Figure 1. The Seifert surface F_Γ for $\Gamma = A_8$, together with its distinguished filling configuration $\mathcal{C}_\Gamma = \{\alpha_1, \dots, \alpha_8\}$ of curves. The curve b is disjoint from $\mathcal{C}_\Gamma \setminus \{\alpha_6\}$ and intersects α_6 once. A regular neighbourhood of b in F_Γ is an unknotted annulus with three full twists.

5.3. Obstructing explicit twists for Type A_n . We illustrate the obstruction in Lemma 5.1 with the following result.

Proposition 5.6. *Fix n even and $n \geq 8$. Let $F = F_n \subset D^4$ be the minimal genus Seifert surface for the $(2, n+1)$ -torus knot with its interior pushed into D^4 , so $F = F_\Gamma$ for $\Gamma = A_{n+1}$. Consider the curve $b \subset F$ depicted in Figure 1 (for $n = 8$). Then $T_{[b]} \in \text{Sp}(H_1(F; \mathbb{Z}_2))[q_F]$ but $T_b \notin \mathcal{E}^{\text{TOP}}(D^4, F)$.*

Proof. By construction $\text{lk}(b, b^+) = -3$, hence $q_F([b]) = 1$, so $T_{[b]} \in \text{Sp}(H_1(F; \mathbb{Z}_2))[q_F]$. Let $\{[\alpha_1], \dots, [\alpha_n]\}$ be the homology classes of the chain of curves depicted in Figure 1 (the homology classes of the curves in $\mathcal{C}_\Gamma$ for $\Gamma = A_n$). We have

$$T_{[b]}([\alpha_i]) = \begin{cases} [\alpha_i], & i \neq 6; \\ [\alpha_1] + [\alpha_3] + [\alpha_5] + [\alpha_6], & i = 6. \end{cases}$$

Consider the free $\mathbb{Z}$ -module with basis $\Delta = \{[\alpha_i]\}_{i=1}^n$ equipped with the Cartan matrix $P(-1) = S^T + S$. Let $W = \langle \text{Ref}_{[\alpha_i]} \rangle$ be the Weyl group of type A_n . Form the root system $\Phi = \bigcup_{w \in W, \alpha \in \Delta} \{w(\alpha)\}$. This consists of all $\alpha \in \mathbb{Z}^n$ with $\alpha \cdot \alpha = -2$, where $\cdot$ is the bilinear form obtained from $P(-1)$. The -2 -vectors in the A_{n+1} -root lattice are exactly the

$$(5.1) \quad \pm(\alpha_i + \alpha_{i+1} + \dots + \alpha_j)$$

for some $j > i$. Any integral matrix w with $w^T P(-1) w = P(-1)$ sends roots to roots, hence the columns of the matrix w take the form (5.1); we call such columns *consecutive interval vectors*.

If T_b extends, then, by Theorem E, $T_{[b]} \equiv w \in W \pmod{2}$ and the column $T_{[b]}\alpha_6$ must be a consecutive interval vector. But $T_{[b]}\alpha_6 \equiv \alpha_1 + \alpha_3 + \alpha_5 + \alpha_6 \pmod{2}$, giving a contradiction. $\square$

Remark 5.7. We comment on similarities between the obstructions arising in the proof of Proposition 5.6 (and Theorem E) and work of A'Campo [A'C79]. A'Campo realized the *monodromy of the singularity* A_{n+1} as the image of a representation $p: B_{n+1} \rightarrow \text{Sp}(H_1(F_n; \mathbb{Z}_2)) \cong \text{Sp}_n(\mathbb{Z}_2)$ of the *braid group* on $(n+1)$ -strands. A'Campo computed $p(B_{n+1}) = W_2(A_{n+1})$ [A'C79, Theorem 1 and Lemma 6]. Another perspective on this obstruction is that it comes from the unitarity of the (reduced) *Burau representation* $B_{n+1} \rightarrow \text{GL}_n(\mathbb{Z}[t, t^{-1}])$, reducing at $t = 1$ and $t = -1$. Our obstructions are obtained in a similar fashion.

5.4. Flexibility after stabilization. If $h \subset M$ is an embedded 3-dimensional 1-handle intersecting a surface $\Sigma \subset M$ only in its attaching region, an *internal stabilization* of Σ replaces $\Sigma \cap h$ with the remaining portion of ∂h (see [BS16] for more details). If the knot group of Σ is cyclic, stabilization is unique to isotopy [BS16, Lemma 3]. In particular, it suffices to add a handle supported in a small 4-ball. We call such a handle attachment *trivial*. Up to isotopy, it is equivalent to taking the relative connected sum $(M, \Sigma) \# (S^4, T)$ where $T \subset S^4$ is an unknotted torus.

Theorem D. *There exist infinitely many non-isotopic pushed in Seifert surfaces in D^4 that are not topologically spin-flexible rel boundary, but become topologically spin-flexible after a trivial internal stabilization.*

Proof. Consider the family of Seifert surfaces F_n arising from the A_{n+1} -singularity for n even and $n \geq 8$ (so $g(F_n) = n/2 \geq 4$). By Lemma 5.6, they are not topologically spin-flexible. Consider the connected sum $F_n \# T$ where $T \subset S^4$ is an unknotted torus. Let Δ denote the surgery 2-disk for this connected sum. We view the interior of $F_n \# T$ as sitting inside an S^3 slice of a collar about ∂D^4 . Let $\Sigma = F_n \# T \cup D$ be the (abstract) genus $g = n/2 + 1$ surface obtained by capping off the boundary of $F_n \# T$ with a disk D . Extending mapping classes of $F_n \# T$ by the identity yields a homomorphism

$$p_D : \text{Mod}(F_n \# T) \rightarrow \text{Mod}(\Sigma).$$

Let $UT(F_n \# T) \rightarrow F_n \# T$ be the unit tangent bundle of $F_n \# T$. The kernel of p_D is the *disk-pushing subgroup* $\pi_1(UT(F_n \# T)) \leq \text{Mod}(F_n \# T)$ (see [FM12, Section 4.2.5]). We have the *disk-pushing sequence*

$$1 \rightarrow \pi_1(UT(F_n \# T)) \rightarrow \text{Mod}(F_n \# T) \rightarrow \text{Mod}(\Sigma) \xrightarrow{p_D} 1.$$

Rochlin's form $q_{F_n \# T}$ extends uniquely to a quadratic form $q : H_1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$. The disk-pushing subgroup acts trivially on the homology of $H_1(F_n \# T; \mathbb{Z}) \cong H_1(\Sigma; \mathbb{Z})$, and so restricting the disk-pushing sequence yields

$$1 \rightarrow \pi_1(UT(F_n \# T)) \rightarrow \text{Mod}(F_n \# T)[q_{F_n \# T}] \xrightarrow{p_D} \text{Mod}(\Sigma)[q] \rightarrow 1.$$

We show

$$p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T)) = \text{Mod}(\Sigma)[q].$$

By Corollary 4.9, $\mathcal{I}(F_n \# T) \leq \mathcal{E}^{\text{TOP}}(D^4, F_n \# T)$. As $\pi_1(UT(F_n \# T)) \leq \mathcal{I}(F_n \# T)$, it will then follow that

$$\mathcal{E}^{\text{TOP}}(D^4, F_n \# T) = \text{Mod}(F_n \# T)[q_{F_n \# T}].$$

We use Lemma 3.11 to prove that $p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T)) = \text{Mod}(\Sigma)[q]$. We must show that $p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T))$ contains an r -spin mapping class group and a genus one bounding pair. The latter is a consequence of Corollary 4.9. For the former, we produce the r -spin lower bound directly, using the finite generating sets provided by Proposition 3.13. Let $\mathcal{C} = \{\gamma_1, \dots, \gamma_n\}$ be the distinguished filling configuration of curves on F_n . We start by expanding the configuration $\mathcal{C}$ to fill $F_n \# T$ and to ensure the existence of an E_6 subgraph. Choose disjoint nonseparating simple closed curves α' and α'' in F_n such that

- (a) α' is disjoint from $\mathcal{C} \setminus \{\gamma_3\}$ and geometrically dual to γ_3 ;
- (b) α'' is disjoint from $\mathcal{C} \setminus \{\gamma_5\}$, geometrically dual to γ_5 , and $\alpha' \cup \gamma_4 \cup \alpha''$ bounds a pair of pants in F_n .

For homological reasons, $q_{F_n}(\alpha')$ or $q_{F_n}(\alpha'') = 1$. Let $\alpha \in \{\alpha', \alpha''\}$ be a (in fact, the) curve with q_{F_n} -value one. The quadratic form q_{F_n} is the mod 2 reduction of the Seifert form on F_n (as is true for any pushed in Seifert surface). Therefore

$$\text{lk}(\alpha, \alpha^+) = 2k + 1$$

for some $k \in \mathbb{Z}$. Choose a simple closed curve $\delta \subset T \setminus \Delta$ with $\text{lk}(\delta, \delta^+) = -2k \in \mathbb{Z}$ and which intersects the standard meridional loop m of $T \setminus \Delta$ once. Let $\gamma = \alpha +_\varepsilon \delta$ be the connected sum of α and δ along an arc ε disjoint from the configuration $\mathcal{C}$. As δ is contained in a ball disjoint from γ ,

$$\text{lk}(\gamma, \gamma^+) = \text{lk}(\alpha, \alpha^+) + \text{lk}(\delta, \delta^+) = 2k + 1 - 2k = 1.$$

Let $\beta = T_\delta(m) \subset T \setminus \Delta$. The curve β is geometrically dual to δ and has Seifert framing $\text{lk}(\beta, \beta^+) = 1 \pmod{2}$, since $q_T(\beta) = q_T(m) + q_T(\delta) + 1 = 1 \pmod{2}$ and the restriction of q_T to $H_1(T \setminus \Delta; \mathbb{Z}/2\mathbb{Z})$ is the mod 2 Seifert form.

Let $\mathcal{D} = \mathcal{C} \cup \{\gamma, \beta\}$. By the Hopf band trick (Proposition 2.7), Dehn twists along curves in $\mathcal{C}$ are (smoothly) extendable. The spin-flexibility of unknots rel boundary (Proposition 2.6) implies that $T_\beta \in \mathcal{E}^{\text{TOP}}(D^4, F_n \# T)$. Corollary 4.20 applied to the curve γ yields

$$\Gamma_{\mathcal{D}} := \langle T_c : c \in \mathcal{D} \rangle \leq \mathcal{E}^{\text{TOP}}(D^4, F_n \# T).$$

By Proposition 3.13,

$$p_D(\Gamma_{\mathcal{D}}) = \text{Mod}(\Sigma)[\psi],$$

where $\text{Mod}(\Sigma)[\psi]$ is the stabilizer of an $n = 2g - 2$ winding number function ψ on Σ . Recall that the winding number function ψ is uniquely determined by specifying that $\psi(\mathcal{D}) = 0$, that is, each curve in $\mathcal{D}$ is admissible. Therefore

$$\text{Mod}(\Sigma)[\psi] \leq p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T)).$$

By Lemma 3.11 and Corollary 4.9,

$$p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T)) = \text{Mod}(\Sigma)[\bar{\psi}],$$

where $\bar{\psi}$ denotes the mod 2 reduction of ψ . By construction, the mod 2 reduction $\bar{\psi}$ of ψ is identified with the form q (they agree on the basis $\mathcal{D}$), that is, $\bar{\psi}(c) + 1 = q(c)$ for any curve $c \subset \Sigma$. It follows that

$$p_D(\mathcal{E}^{\text{TOP}}(D^4, F_n \# T)) = \text{Mod}(\Sigma)[q].$$

□

6. FLEXIBILITY WITHIN r -SPIN LINEAR SYSTEMS AND FAKE VANISHING CYCLES

Let X be a smooth projective surface and $L \in \text{Pic}(X)$ a holomorphic line bundle on X .

6.1. Monodromy of linear systems and higher spin structures. Let $|L| := \mathbb{P}H^0(X, L)$. Recall that elements of $|L|$ may be identified with their zero loci in X (that is, two sections have the same zero locus if and only if they are scalar multiples of one another). Let $U_L \subset |L|$ denote the open subset of $|L|$ consisting of complex curves in X that are smooth in the algebro-geometric sense (the complement of the *discriminant locus*). The tautological family $\mathcal{C}_L$ over U_L is defined by

$$\mathcal{C}_L = \{(C, x) \mid C \in U_L, x \in C\}.$$

The natural projection $\pi_L : \mathcal{C}_L \rightarrow U_L$ equips $\mathcal{C}_L$ with the structure of a smooth fiber bundle over U_L . The fibers of π_L are the complex curves in U_L . In particular, after fixing a $C \in U_L$ there is a *monodromy representation*

$$\rho_L : \pi_1(U_L, C) \rightarrow \text{Mod}(C).$$

Geometrically, a loop γ in U_L based at C gives a one-parameter family of embedded (real) surfaces in the 4-manifold X . By isotopy extension, this determines a diffeomorphism of the pair (X, C) whose restriction to C determines the mapping class $\rho_L(\gamma)$.

Definition 6.1. Let $\Gamma_L = \text{im}(\rho_L)$. We refer to Γ_L as the *monodromy of the linear system* $|L|$.

Corollary 6.2. For a smooth curve $C \in |L|$,

$$\Gamma_L \leq \mathcal{E}^{\text{DIFF}}(X, C).$$

In general, the monodromy of a linear system is governed by a *higher spin structure* on C . Fix an r -th root $E \in \text{Pic}(X)$ of the adjoint line bundle of L , that is, $E^{\otimes r} \cong K_X \otimes L$. As X is simply connected, the first Chern class gives a monomorphism $c_1 : \text{Pic}(X) \rightarrow H^2(X; \mathbb{Z})$ [GH94] and $H^2(X; \mathbb{Z})$ is torsion free. In particular, the r -th root E of $K_X \otimes L$ is unique. Adjunction gives an isomorphism $(K_X \otimes L)|_C \cong K_C$. In particular, we have a fiberwise branched covering of the holomorphic tangent bundle T_C of C ,

$$E^{-1}|_C \rightarrow (E^{-1})|_C^{\otimes r} \cong T_C,$$

on fibers given by $z \rightarrow z^r$. This determines a cohomology class $\xi_E \in H^1(UTC; \mathbb{Z}/r\mathbb{Z})$ on the unit tangent bundle UTC of C . By Theorem 3.5, ξ_E determines a winding number function

$$\psi_E : \mathcal{S}(C) \rightarrow \mathbb{Z}/r\mathbb{Z}.$$

The monodromy of the linear system $|L|$ must stabilize this winding number function.

Proposition 6.3 ([BS25, Lemma 3.3]). *An r -th root E of the adjoint line bundle $K_X \otimes L$ determines a winding number function $\psi_E : \mathcal{S}(C) \rightarrow \mathbb{Z}/r\mathbb{Z}$, and*

$$\Gamma_L \leq \text{Mod}(C)[\psi_E].$$

Definition 6.4. A linear system $|L|$ has r -spin *monodromy* if, for a maximal root E of the adjoint line bundle $K_X \otimes L$, we have

$$\Gamma_L = \text{Mod}(C)[\psi_E].$$

Banerjee and Salter proved that if $\pi_1 X = 1$ and L is *sufficiently ample*, the system $|L|$ has r -spin monodromy [BS25, Theorem A]. Their result is essential in proving Theorem A.

6.2. k -jet ample line bundles. A line bundle L is 0-jet ample if and only if it is *basepoint free* and 1-jet ample if and only if it is *very ample*. For a precise definition of k -jet *ampleness*, we refer the reader to [BS93]. For our purposes, k -jet ampleness of L allows us to find a section of L with any prescribed k -jet at a point $x \in X$. We make use of the following property when establishing that a bundle is k -jet ample.

Proposition 6.5. *If L_1 is k_1 -jet ample, and L_2 is k_2 -jet ample, then $L_1 \otimes L_2$ is $(k_1 + k_2)$ -jet ample.*

Proof. See [BS93, Lemma 2.2]. □

Lemma 6.6. *Assume that $L \in \text{Pic}(X)$ is 6-jet ample. Then there exists a smooth curve $C \in |L|$ and a closed ball $B \subset X$ such that $C \cap B$ is a $\mathbb{Z}$ -surface in B of genus three.*

Proof. Fix a point $p \in X$ with coordinate neighbourhood $U \cong \mathbb{C}^2$ and local (analytic) coordinates x, y . Let $\varphi(x, y) = x^3 + y^4 \in \mathbb{C}[x, y]$. Then there is a section $f_0 \in H^0(X; L)$ such that $Z(f_0)$ has the singularity type of φ at p and is smooth everywhere else [BS25, Lemma 5.7]. A well chosen smoothing of f_0 does not change the isotopy type of $Z(f_0)$ outside of a small closed ball B about $p \in X$, but replaces $Z(f_0) \cap B$ with a Milnor fiber Σ_φ of the singularity φ . By Corollary 5.2, the Milnor fiber of the singularity φ is a $\mathbb{Z}$ -surface in B . Moreover, up to isotopy, it is obtained by pushing the interior of a minimal genus Seifert surface for the $(3, 4)$ -torus knot into the interior of the ball B , so Σ_φ has genus three. □

Remark 6.7. Lemma 6.6 can be extracted from the literature as follows. Replace [Ban25, Proposition 7] in the proof of [Ban25, Lemma 1] with [BS25, Lemma 5.7]. This follows the program suggested at the end of first section in [Ban25]. One replaces working with sufficiently large powers of an ample line bundle with sufficient ampleness conditions.

6.3. Topological flexibility within linear systems. We now prove the main result of this section.

Theorem 6.8. *Let X be a projective surface with $\pi_1(X) = 1$. Let $L \in \text{Pic}(X)$ be 6-jet ample. Assume that $|L|$ has r -spin monodromy (Definition 6.4). Then smooth curves in the linear system $\mathbb{P}H^0(X; L)$ are topologically flexible if r is odd, and topologically spin-flexible if r is even.*

Proof. As any two smooth curves in $|L| = \mathbb{P}H^0(X; L)$ are isotopic in X , it suffices to prove the claim for a single smooth curve in $|L|$. By Lemma 6.6 we can find a smooth curve $C \in |L|$ and an embedded 4-ball $B \subset X$ such that $\Sigma := C \cap B$ is a properly embedded $\mathbb{Z}$ -surface in B of genus three. By Corollary 4.9,

$$\mathcal{I}(\Sigma) \leq \mathcal{E}^{\text{TOP}}(B, \Sigma).$$

Let $T_a T_b^{-1} \in \mathcal{I}(\Sigma)$ be a genus one bounding pair on Σ . Let $\psi_E : \mathcal{S}(C) \rightarrow \mathbb{Z}/r\mathbb{Z}$ be the mod r winding number function determined by the maximal root E of the adjoint line bundle $K_X \otimes L$. By assumption, $\text{Mod}(C)[\psi_E] \leq \mathcal{E}^{\text{DIFF}}(X, C)$. Therefore

$$\langle \text{Mod}(C)[\psi_E], T_a T_b^{-1} \rangle \leq \mathcal{E}^{\text{TOP}}(X, C).$$

Assume r is odd. By Lemma 3.11,

$$\langle \text{Mod}(C)[\psi_E], T_a T_b^{-1} \rangle = \text{Mod}(C)$$

and so C is topologically flexible.

Assume r is even. Libgober [Lib88] identifies the mod 2 reduction $\overline{\psi_E}$ of ψ_E as the stabilizer of Rochlin's quadratic form $q_C : H_1(C; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$. By Lemma 3.11,

$$\text{Mod}(C)[q_C] \leq \mathcal{E}^{\text{TOP}}(X, C).$$

On the other hand, by Corollary 2.4, $\mathcal{E}^{\text{TOP}}(X, C) \leq \text{Mod}(C)[q_C]$, and hence C is topologically spin-flexible. $\square$

Remark 6.9 (Recovering Theorem A). If $L = L_1 \otimes L_2$ for L_1 very ample (1-jet ample) and L_2 6-jet ample, then [BS25, Theorem A] asserts that $|L|$ has r -spin monodromy. The bundle L is 7-jet ample, so 6-jet ample. Now apply Theorem 6.8.

Corollary B. *Smooth plane curves in $\mathbb{CP}^2$ are topologically (spin) flexible. Smooth curves in $\mathbb{CP}^1 \times \mathbb{CP}^1$ of bidegree (d_1, d_2) with $d_1, d_2 \geq 6$ are topologically (spin) flexible.*

Proof. For plane curves of degree at most five, smooth flexibility follows from work of Hirose (for cubics and quartics) [Hir05] and Salter (for quintics) [Sal25]. In the remaining cases, we apply Theorem 6.8.

For $d \geq 6$, smooth degree d plane curves are the zero loci of sections of $\mathcal{O}_{\mathbb{CP}^2}(d) = \mathcal{O}_{\mathbb{CP}^2}(1)^{\otimes d}$, and the line bundle $\mathcal{O}_{\mathbb{CP}^2}(d)$ is d -jet ample. It follows from [Sal19, Theorem 1.3] and [CS21, Corollary 1.3] that $\mathcal{O}_{\mathbb{CP}^2}(d)$ has r -spin monodromy.

A smooth curve $C_{d_1, d_2} \subset \mathbb{CP}^1 \times \mathbb{CP}^1$ of bidegree (d_1, d_2) is the zero locus of a smooth section of $L_{d_1, d_2} = \text{pr}_1^* \mathcal{O}_{\mathbb{CP}^1}(d_1) \otimes \text{pr}_2^* \mathcal{O}_{\mathbb{CP}^1}(d_2)$. The line bundle L_{d_1, d_2} is k -jet ample if and only if $d_1 \geq k$ and $d_2 \geq k$ [BS93, Proposition 5.1]. In particular, if $d_1, d_2 \geq 6$, the bundle L_{d_1, d_2} is 6-jet ample. By [Sal19, Theorem A] and [CS21, Corollary 1.3], L_{d_1, d_2} has r -spin monodromy. The claim now follows from Theorem 6.8. $\square$

6.4. Fake vanishing cycles. Let $\mathbb{D} \subset \mathbb{C}$ be the closed unit disk. Recall that a *nodal degeneration* of a smooth curve $C \in |L|$ consists of a family of curves $\{C_t\}_{t \in \mathbb{D}} \subset X$ parameterized by $\mathbb{D}$ such that C_0 has a single nodal singularity at $p \in C_0$, each C_t is smooth for $t \neq 0$, and $C_1 = C$. A simple closed curve $a \subset C$ is called a *vanishing cycle* if under parallel transport to C_0 within the family $\{C_t\}_{t \in \mathbb{D}}$ it maps to the point p . Under such an isotopy, the curve $a \subset C$ traces out a smoothly embedded disk $D_a \subset X$ with relative Euler number $e_C(D_a) = -1$ (see [GS99]). We make the following definition in the topological category.

Definition 6.10. Let $\Sigma \subset M$ be a properly embedded surface in a topological 4-manifold M . A simple closed curve $a \subset \Sigma$ is a *topological vanishing cycle* if it bounds a topologically embedded disk $D \subset X$ with interior disjoint from Σ and relative Euler number $e_\Sigma(D) = \pm 1$.

The terminology in Definition 6.10 is motivated by Proposition 2.7. Indeed, if $\gamma \subset C$ is a vanishing cycle, the associated nodal degeneration can be used to realize the (right handed) Dehn twist about γ as an extendable mapping class.

Definition 6.11. A simple closed curve $a \subset C \in |L|$ is called a *fake vanishing cycle* if it is a topological vanishing cycle but not a vanishing cycle on C .

We now use Theorem A to establish the existence of fake vanishing cycles. We need the following result which, for r -spin linear systems, characterizes vanishing cycles as admissible curves.

Proposition 6.12 ([BS25, Corollary 1.9]). *Let $L \in \text{Pic}(X)$ and $C \in |L|$ a smooth curve. Let ϕ_E be the winding number function determined by a maximal root of the adjoint line bundle. Assume that $|L|$ has r -spin monodromy, that is, $\Gamma_L = \text{Mod}(C)[\phi_E]$. Then a nonseparating simple closed curve $a \subset C$ is a vanishing cycle if and only if $\phi_E(a) = 0$.*

Corollary C. *Let X be a projective surface with $\pi_1(X) = 1$. Let $L \in \text{Pic}(X)$ satisfy the hypothesis of Theorem A. Fix a smooth curve $C \in |L|$. If the maximal root of the adjoint line bundle $K_X \otimes L$ in $\text{Pic}(X)$ has order at least three, then there exists a fake vanishing cycle $a \subset C$.*

Proof. By [BS25, Theorem A], the linear system $|L|$ has r -spin monodromy, where r is the maximal root of the adjoint line bundle $K_X \otimes L$. By assumption, $r \neq 1, 2$. Let E be a maximal root of $K_X \otimes L$ and fix $C \in |L|$ a smooth curve. Let $\phi_E: \mathcal{S}(C) \rightarrow \mathbb{Z}/r\mathbb{Z}$ be the associated winding number function. As L is 6-jet ample, the genus of C is at least three (Lemma 6.6). By the r -spin change of coordinates principle (requiring genus at least two, see [CS21, Section 5.2]), there exists a nonseparating simple closed curve $a \subset C$ with $\phi_E(a) \neq 0 \in \mathbb{Z}/r\mathbb{Z}$. If r is even, we can arrange for

$\phi_E(a) = 0 \pmod{2}$. Let $c \subset C$ be an admissible curve, that is, $\phi_E(c) = 0$. By Proposition 6.12, c is a vanishing cycle. By Theorem 6.8, C is topologically (spin) flexible. In particular, there exists an orientation preserving homeomorphism of pairs $f : (X, C) \rightarrow (X, C)$ such that $f(c) = a$. This realizes the curve $a \subset C$ as a fake vanishing cycle. $\square$

Remark 6.13. The proof of Corollary C shows that, if r is odd, any nonseparating non-admissible curve is a fake vanishing cycle. If r is even (so C is characteristic), any curve $a \subset C$ with $q_C([a]) = 1$ is a fake vanishing cycle. If $r = 1, 2$ and the linear system $|L|$ has r -spin monodromy, fake vanishing cycles do not exist. For example, if $C_d \subset \mathbb{CP}^2$ is a smooth curve of degree $d \leq 5$, fake vanishing cycles do not exist on C_d , but they do exist if $d \geq 6$. By Proposition 2.7, we see that a curve with r -spin monodromy is topologically (spin) flexible if and only if there exist fake vanishing cycles.

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DEPT OF MATHEMATICS; UNIV OF NOTRE DAME; HURLEY HALL; NOTRE DAME, IN 46556; USA
 SCHOOL OF MATHEMATICS, UNIVERSITY OF BRISTOL

Email address: jlehman4@nd.edu
 et24788@bristol.ac.uk